



Applied Microeconometrics

Fundamentals

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Disclaimer: Sections and lines in brown are ‘under construction’.

Motivation

Research questions related to the goal of sustainable development bring together social and natural systems, and are thereby particularly conducive to interdisciplinary work. The social system part demands some training in the social sciences, and in effect interdisciplinary researchers may have an economics background.

Applied microeconomics work in recent years has largely concerned the identification of causal relationships between variables, such that the current dominant methods and terminology are largely fitted to that goal. In applied work from other disciplines, one is likely to encounter alternative types of models, estimation methods, terminology, and even ultimate goals of the statistical analysis (e.g., predictive inference vs causal inference). If nothing else, an applied interdisciplinary researcher should be able to communicate with these different academic disciplines. This means notably understanding what a given method does in statistical terms, in other words: where it fits in the ‘family tree’ of statistical approaches. This will enable them to both: choose the most appropriate method given the problem at hand (when understanding what the method is doing, the empowered researcher need not resort only to the most common method in a given discipline), and justify that choice in front of the different disciplinary communities.

The purpose of this document is therefore twofold:

1. To detail the typical methods of applied microeconomics, which are our reference base. This includes defining and distinguishing common notions that may be conflated (*a model, an equation, a regression, a specification, an estimation method...*);
2. To put those into context, i.e., place them in the greater ‘family tree’ or space of statistical methods, and delineate a few other branches of that tree that may be relevant for empirical interdisciplinary research.

Let us start by defining microeconometrics:

Econometrics = (originally) the application of statistical methods to economic data, in order to quantify the relationships postulated by economic theory and estimate the parameters of economic models.

Microeconometrics = the use of these statistical methods to study microdata pertaining to individuals, households, and firms.

Ultimately, applied economics is an area of applied statistics informed by economic theory.

1 Data

Empirical studies can be separated into two classes, based on the type of data collected:

Study	Data collection
Experimental	The researcher records data about subjects while applying treatments and controlling conditions (active participation).
Observational	The researcher records data about subjects without applying a treatment (passive participation). If the goal is to uncover characteristics of a population, they may: • inspect the entire population: perform a census; • inspect a subset: take sample data from the population probability distribution.

Types of observational data Observational data can belong to specific categories, based on the dimensions of units (L) and time (T):

- **Cross-sectional [L]**: observations for several units, at one point in time;
- **Time series [T]**: observations for a single unit, at repeated points in time;
- **Longitudinal [L × T]**: observations for several units, at repeated points in time.

When *the same units* are observed over time, we have **panel data**.¹ The panel can be:

- balanced: all observed units $\{i\}$ have data across all periods $\{t\}$;
- unbalanced: some units have more observations than others.

Variation *between* units at one point in time is called *between*-variation, while variation *within* one unit across time is called *within*-variation. The total variance of observed variables can be split into within- and between-variation.

One of the strengths of longitudinal data is their potential for estimating causal relationships, as they provide a way to deal with observable and unobservable effects.

Sampling procedures

Random sampling ensures that the *data* probability distribution is the same as the *population* distribution. If sampling isn't random, it is **biased**: the data distribution differs from the population distribution.

Common random sampling procedures include:

- **Simple random sampling** — the assumption on which statistical inference theory is based.
- **Stratified random sampling**: the population is divided into S subgroups or “strata”, of $N_1 \neq N_2, \dots, N_S$ units. Simple random samples of sizes $n_1, n_2, \dots, n_S$ are drawn independently.
 - **Proportionate stratified random sampling**. Ex: in a “10% sample, stratified across subgroups”, the same fraction is applied on each subgroup.

¹“Panel data” and “longitudinal data” are often used interchangeably, as it is often the same units that are observed over time. However keeping the distinction, as delineated in [Mertens et al. \(2017\)](#), can be useful.

2 Statistical models

A model is a formal representation of a theory about a system, used to describe, explain, or predict that system.

A statistical model is a mathematical model of the data generating process (DGP) that generates the random sample $\{(Y_i, X_i)\}_{i=1}^n$, of which we observe one realization $\{(y_i, x_i)\}_{i=1}^n$.^a

- What distinguishes it from other mathematical models is that it has a stochastic component: some quantities are *random* variables, and therefore have probability distributions.^b
- It is written as relationships between these random variables, some fixed quantities, and unknown parameters, and is used to study the distribution and variation of the random variables.

^aFormally, it combines the set of possible observations, i.e., sample space, $\mathcal{S}$, and a collection $\mathcal{P}$ of joint probability distributions on $\mathcal{S}$ (which ideally would include the “true” probability distribution induced by the DGP; but it doesn’t need to, we accept that all models are false).

^bIndeed, the task of statistics can be described as quantifying evidence and reasoning *under uncertainty*.

2.1 Notations

For statistical analysis, we adopt a view of the world in three levels:

- (1) the generic-unit population level or DGP, where objects such as Y and X are random variables;
- (2) the random-sample level, which is a sample from that population, i.e., a finite set of random variables generated by the DGP, such as $\{(Y_i, X_i)\}_{i=1}^n$,²
- (3) the realized-sample level, i.e., the variables’ realized values, such as $\{(y_i, x_i)\}_{i=1}^n$.

Formally, a statistical model specifies a probability distribution, or a family of probability distributions, for random variables. Its objects are random variables. One may also write the corresponding realized-sample equation by replacing the random variables with their realized values, but this equation is not itself the statistical model.

For clarity, in this document, we will strive to use the following notations: latin characters for variables and greek characters for parameters; upper case letters for random variables and lower cases for their realized values (with the exception of disturbances and residuals, which will always be written as lower case letters); italic letters for scalars, roman letters for vectors, and sans serif letters for matrices. For example:

- Y denotes the random outcome of a generic population unit, Y_i the random outcome of unit i , $Y = (Y_1, \dots, Y_n)'$ the random outcome vector of all units;
- X_{ij} denotes the j^{th} random covariate of unit i , $X_i = (X_1, \dots, X_k)'$ the $k \times 1$ random covariate vector of unit i , $X = (X'_1, \dots, X'_n)'$ the $n \times k$ matrix of all random covariates of all units;
- u_i denotes the random disturbance of unit i , $u = (u_1, \dots, u_n)'$ the random disturbance vector;
- $y, y_i, y, x_{ij}, x_i, x, e_i, e$ denote the realizations of $Y, Y_i, Y, X_{ij}, X_i, X, u_i, u$, respectively.

2.2 Microeconometrics models

Empirical investigations in microeconometrics aim to uncover important relationships to understand microeconomic behavior. They can broadly be separated into two types of approaches, depending on the extent to which they rely on microeconomic theory:

- **Structural analysis** heavily depends on economic theory. Model specifications are derived from specifications of the economic behavior. The goal is to analyze structural relationships between interdependent microeconomic variables (e.g., to estimate structural parameters that characterize individual preferences or technological relationships).

$$g(Y, X, u; \xi) = 0, \quad \xi = \text{structural parameters}$$

² Δ In the random sample $\{(Y_i, X_i)\}_{i=1}^n$, each Y_i is an individual random variable. A common mistake is to believe $Y_1, \dots, Y_n$ are different realizations of a single random variable Y . Instead, we are dealing with a *sequence* of n random variables. If for example the generic Y represents a person’s height, and there is a sample of n persons, then $Y_{i=1}$ is the random variable representing the height of the first person in sample, $Y_{i=2}$ is the random variable representing the height of the second person in sample, etc. Each Y_i of the length- n sequence will then have a realization y_i which we may or may not observe.

- **Reduced-form analysis** makes much less use of economic theory. The goal is to uncover associations among variables, by using regression models.

The **reduced form** of a system of structural equations is the result of solving the system for the dependent (i.e., nonlagged and endogenous) variables. This **gives the dependent variables as functions of the independent variables** (exogenous variables or lags of the dependent).

$$Y = h(X, v; \pi), \quad \pi = \text{reduced-form parameters that are functions of } \xi$$

2.3 ⊃ Regression models

A **regression model** is a statistical model for the **conditional distribution** $Y|X$ of an outcome random variable Y given covariates X , or for some feature(s) of that distribution.

The random variables $\{Y, X_1, \dots, X_k\}$ have an unknown joint distribution. Rather than modeling that full joint distribution, regression models focus on the *conditional distribution* of Y given X . Some models specify the entire conditional distribution, others specify only features $\phi()$ of it—e.g., moments or functionals. Examples:

Regression model	Distributional specification	Feature specification
	$Y X \sim \mathcal{D}(m(X; \beta), \eta)^3$	$\phi(Y X) = m(X; \beta)$
normal classical linear	$Y X \sim \mathcal{N}(X'\beta, \sigma^2)$	$\mathbb{E}[Y X] = X'\beta; \mathbb{V}[Y X] = \sigma^2$
classical linear		$\mathbb{E}[Y X] = X'\beta; \mathbb{V}[Y X] = \sigma^2$
general linear		$\mathbb{E}[Y X] = X'\beta$
logistic	$Y X \sim \text{Ber}(\text{logit}^{-1}(X'\beta))$	$\begin{cases} \mathbb{E}[Y X] = \text{logit}^{-1}(X'\beta); \\ \mathbb{V}[Y X] = \text{logit}^{-1}(X'\beta)(1 - \text{logit}^{-1}(X'\beta)) \end{cases}$
Poisson	$Y X \sim \text{Poisson}(e^{X'\beta})$	$\mathbb{E}[Y X] = e^{X'\beta}; \mathbb{V}[Y X] = e^{X'\beta}$
negative binomial	$Y X \sim \text{NB}(e^{X'\beta}, r)$	$\mathbb{E}[Y X] = e^{X'\beta}; \mathbb{V}[Y X] = e^{X'\beta} + e^{2X'\beta}/r$
quantile		$\mathbb{Q}[Y X](\alpha) = X'\beta_\alpha$
Cox proportional hazard		$h(t X) = h_0(t)e^{X'\beta}$

Writing a regression model implies that we consider that a random sample $\{(Y_i, X_i)\}_{i=1}^n$ is generated by the process described by that model. The model can be written:

- at the generic-unit population level: $Y|X \sim \mathcal{D}(m(X; \beta), \eta)$.
 $\triangle$ *By itself, this model says nothing about potential dependence across units.*⁴
- at the random sample level and stacked—i.e., in vector form: $Y|X \sim \mathcal{D}_n(m(X; \beta), \eta)$,⁵ where $\mathcal{D}_n$ incorporates the joint (cross-unit) conditional dependence structure in Y , if any.
- at the random sample level and individualized—i.e., as a system of n equations: $Y_i|X_i \sim \mathcal{D}(m(X_i; \beta), \eta)$, $i = 1, \dots, n$, if supplemented with the joint conditional dependence assumptions.

Choosing a model specification To carry out regression analysis, one must first choose a model specification: select which independent variables to include and an appropriate functional form of the relation between Y and X . Specification error occurs when either the functional form or the choice of independent

³Where $\mathcal{D}$ denotes a family of conditional distributions, $m(X; \beta)$ the regression function, and η are unmodeled distribution parameters. Note also that in the context of Bayesian inference, as parameters are considered random variables, one may encounter a DGP written conditional on β : $m(X|\beta)$. In frequentist inference, β is considered fixed and so is not conditioned on.

⁴To describe dependence at the population level, the population object would have to be a collection of jointly distributed random variables $\{(Y_i, X_i)\} : i \in \mathcal{I}$ (rather than merely one generic pair (Y, X)) whose joint distribution would *not* factorize into marginal distributions: $F_{Y|X}(y|X) \neq \prod_{i=1}^n F_{Y_i|X}(y_i|X)$.

⁵ $m()$ is the vector-valued extension of the scalar-valued $m()$, obtained by stacking the individual regression functions: $m(X; \beta) = (m(X_1; \beta), \dots, m(X_n; \beta))'$.

variables poorly represents relevant aspects of the true DGP. Though “correct specification” is, in practice, unrealistic, as we do not observe the true DGP, we try to avoid the three basic types of misspecification:

- using an inappropriate functional form;
- including an X that is theoretically irrelevant (it has no partial effect on Y) $\rightarrow$ *overspecified* model;
- excluding an X that is theoretically relevant (it may cause Y) $\rightarrow$ *underspecified* model.

▷ **Non-/Semi-/Parametric models** The specification of a statistical model can be:

- **parametric or finite-dimensional:** the model is a family of probability distributions that has a fixed *finite* number of parameters.⁶ We assume that the population distribution belongs to this family for some unknown value of θ .
 - For *regression* models, this means that the conditional distribution of Y given X is specified up to finitely many parameters: $Y|X \sim F_{Y|X}(\cdot | x; \theta)$.
 - The parameters are said to be “**identifiable**” when distinct parameter values imply distinct distributions.⁷ Equivalently: *if the population distribution were known, could we uniquely determine the parameter?*
 - * The model being “well-identified,” i.e., the identification of all its parameters, is necessary, though not sufficient, for consistent estimation—and thus for meaningful statistical inference. It may follow from functional-form or distributional restrictions, or from exclusion, inequality, or moment (e.g., covariance) restrictions.
 - * *Ex. of non-identification: In the linear regression model, perfect collinearity among the regressors means that β isn’t identifiable.*
- **non-parametric or infinite-dimensional:** the model makes no assumptions about a parametric distribution, it determines it from data.⁸ The model has parameters, but their number and nature aren’t fixed in advance.

Ex: Kriging; kernel regression; local polynomial regression, including LOESS.

 - For *regression* models, it means that no finite-dimensional parametric form is imposed on the regression function m .
- **semiparametric:** the model contains both a finite-dimensional set of parameters of interest and an infinite-dimensional nuisance component.

Ex: In the general linear model, β is finite-dimensional, while the conditional distribution of the disturbance $u|X$ is left unspecified: $Y = X'\beta + u$, $\mathbb{E}[u|X] = 0$.

Why care about parametrization? It determines which probability distributions are admitted by the model and which features of the population distribution can be recovered. Parametric models impose stronger restrictions and generally permit more precise estimation, whereas nonparametric and semiparametric models allow greater flexibility at the cost of weaker identification or slower estimation.

⁶Recall that a statistical model is a collection $\mathcal{P}$ of joint probability distributions on some sample space $\mathcal{S}$. We can write it as $\mathcal{P} = \{P_\theta | \theta \in \Theta\}$, where Θ is the parameter space. Hence we can write a parametric model as $\mathcal{P} = \{P_\theta | \theta \in \Theta \subseteq \mathbb{R}^k\}$. Examples of probability distributions with a finite-dimensional parameter vector: the Poisson family of distributions is parametrized by a single number $\lambda > 0$; the normal family is parametrized by two numbers $\{\mu, \sigma\}$.

⁷I.e., the correspondence of each distribution in $\mathcal{P}$ with a θ is 1-1, s.t. $P_{\theta_1} = P_{\theta_2} \Rightarrow \theta_1 = \theta_2$.

⁸Nonparametric regression requires larger sample sizes than regression based on parametric models, because the data must supply the model structure in addition to the model estimates. Nonparametric models also commonly impose other assumptions, such as smoothness, continuity, monotonicity, and strong restrictions/independencies.

3 Statistical inference [under a frequentist approach]

The literature sometimes adopts a broad definition of inference which includes prediction, namely: statistical inference as the use of a statistical model to make generalizations from particular data.⁹ A more restrictive definition of inference instead separates it from prediction: inference uses the model to learn about the data generating process, while prediction uses the model to predict the outcomes for new data points.

In any case, they both contrast with descriptive statistics, which is solely concerned with properties of the observed data—not of a larger population. And they both first require a model that describes the relationship between the variables in the data. In this document we generally use the restrictive terminology that separates prediction from inference as it is most common.

Inferential statistics or **statistical inference** consists in *inferring* properties of a population,^a by calculating statistics from a sample of data drawn from that population.

It combines data and prior assumptions, and generally involves:

- *Estimation*: of values of parameters θ that characterize some feature of interest in the population;
- *Hypothesis testing*: testing for specific values of the unknown parameters θ .

^a*Population*, *DGP*, and *underlying probability distribution* could be used interchangeably. The data observed are of random variables, and we want to estimate parameters θ of their joint probability distribution. Making statistical inferences = deducing properties of probability distributions.

3.1 Frequentist vs Bayesian inference

Consider θ a true, unknown, population quantity of interest. There are two main paradigms for inference, whose difference is rooted in how they interpret probability and represent uncertainty about such a θ .

Frequentist paradigm	Bayesian paradigm
Definition of <i>probability</i> P	
P := the frequency of occurrence of an event; hence only repeatable events have P's (ex: coin flips).	P := one's belief in an event; hence any event, incl. non-repeatable, can have a P.
Implication regarding θ	
$\implies \theta$ is <i>fixed</i> . We can't assign P's to statements such as $\theta \leq \hat{\theta}$. Uncertainty about θ is quantified through the sampling properties of estimators and inferential procedures.	$\implies$ Uncertainty about θ is represented through a <i>random variable</i> Θ , assigned a probability distribution, with possible realizations informally denoted by θ .
Inference about θ using data	
1. Define an estimator $\hat{\theta}_n$ 2. Observe sample data and compute the estimate $\hat{\theta}_n^{\text{obs}}$ 3. Use the sampling distribution of $\hat{\theta}_n$ to construct confidence intervals and hypothesis tests Result: a point estimate and measures of sampling uncertainty, e.g., in the form of a confidence interval, expected to cover the true value ...% of the time. ("time" = number of possible samples from the pop.)	1. Specify a prior P distribution for Θ 2. Observe sample data and update this distribution, by applying Bayes' theorem: $\Pr(\theta \text{data}) = \frac{\Pr(\text{data} \theta) \times \Pr(\theta)}{\Pr(\text{data})}$ Result: a <i>posterior</i> P distribution for Θ . We can compute a 95% credible interval s.t. "after seeing the data, there is a 95% probability that Θ lies in the interval."
Prediction	
Use the realized point estimate $\hat{\theta}_n^{\text{obs}}$ to produce point predictions and prediction intervals.	Use the posterior P distribution of Θ to generate a posterior predictive distribution of the outcome.

⁹Indeed, they need not be that separated. Both consist in making generalizations from data. And all challenges of statistical inference (the ubiquitous problem of generalizing from sample to population; the causal inference problem of generalizing from treatment to control group...) can be framed as problems of prediction (for new individuals outside the sample, of future outcomes under different assigned treatments...) (Gelman et al., 2020, p.3).

The sections below describe the *ABC* of statistical inference in the context of regression analysis, and under the frequentist paradigm, which is most common in econometrics.

3.2 Estimation

3.2.1 Extremum estimators

Consider a set of n observations $\{(y_i, x_i)\}_{i=1}^n$, i.e., realizations of the random variables $\{(Y_i, X_i)\}_{i=1}^n$.

An **estimand** θ is a fixed unknown **population** quantity of interest that we want to estimate, e.g., a parameter or a function of the population distribution. *Ex: the population mean μ_X .*

An **estimator** $\hat{\theta}_n$ of an estimand is a statistic, i.e., a function of the random sample (and therefore a random variable itself): $\hat{\theta}_n = t(X_1, \dots, X_n)$. Its realization will vary sample to sample.

Ex: the sample mean $\bar{X}_n$ is an estimator for the population mean μ_X .

An **estimate** $\hat{\theta}_n^{\text{obs}}$ is a realization of that r.v. $\hat{\theta}_n$, calculated for our specific sample: $\hat{\theta}_n^{\text{obs}} = t(x_1, \dots, x_n)$.

The most common estimators in microeconometrics are extremum estimators: they solve a min/max problem. Consider a regression model of $Y_i|X_i$ with regression function $m(X_i; \beta)$, $\forall i = 1, \dots, n$.

- **Maximum Likelihood (ML)**¹⁰

We want to find the parameter value that makes the observed data most likely *under the assumed model*. The likelihood function $\mathcal{L}$ for a regression model is the probability density/mass function $f_{Y|X}$ evaluated at the observed realizations: $\mathcal{L}(y|x; \theta) := f_{Y|X}(y|x; \theta)$. The ML estimator and estimate are, respectively:

$$\hat{\theta}_{\text{ML}} := \arg \max_{\theta} \ln f_{Y|X}(Y|X; \theta), \quad \hat{\theta}_{\text{ML}}^{\text{obs}} := \arg \max_{\theta} \ln f_{Y|X}(y|x; \theta) = \arg \max_{\theta} \ln \mathcal{L}(y|x; \theta)$$

If observations are conditionally independent, we further have $f_{Y|X}(Y|X; \theta) = \prod_{i=1}^n f_{Y_i|X}(Y_i|X; \theta)$, hence $\hat{\theta}_{\text{ML}} = \arg \max_{\theta} \sum_{i=1}^n \log f_{Y_i|X}(Y_i|X; \theta)$.

- **Least Squares (LS)**

We want to find the parameter value that minimizes the residuals—that is, the discrepancies between the outcomes and the regression function $r := Y - \hat{Y} = Y - m(X; \hat{\beta})$ ¹¹—and more specifically the squared residuals (potentially after some factor adjustment). I.e., we consider a quadratic loss function $Q(v) := v'Wv$, where W is a symmetric positive definite weighting matrix:

$$\hat{\beta}_{\text{LS}} := \arg \min_{\beta} r'Wr$$

The optimal predictor is the conditional mean: $m(X; \beta_{\text{LS}}) := \arg \min_{m(\cdot)} \mathbb{E}[r'Wr] = \dots = \mathbb{E}[Y|X]$.

Only when the model is linear, i.e., linear in the parameters s.t. $m(X; \beta) = X'\beta$, in which case the optimization is called **linear least squares (LLS)**, does the estimator generally have a closed-form solution.¹²

¹⁰Note that the ML estimator is just a type of statistic, and can be conceptualized under either of the frequentist or the Bayesian paradigms. From the vantage point of Bayesian inference, ML is a special case of ‘maximum a posteriori’ estimation that assumes a uniform prior distribution of the parameters. In frequentist inference, ML is a type of extremum estimation, where the objective function is the likelihood.

¹¹We distinguish the *disturbances* or *errors* $u := Y - m(X; \beta)$ which are the unobserved random variables in the population model, i.e., defined for the true unknown value β , and the *residuals* $r(\hat{\beta}) := Y - m(X; \hat{\beta})$, which are the analogue for a given candidate estimator $\hat{\beta}$.

¹²Models such as logistic, Poisson, or negative-binomial regression, which aren’t linear in the parameters (as $m(X; \beta) = \text{logit}^{-1}(X'\beta)$ or $e^{X'\beta}$), can in principle be estimated by *nonlinear* LS—in which case estimators have no closed-form solution and so are computed through numerical optimization—but are conventionally estimated by ML or quasi-ML.

* **Ordinary least squares (OLS)**

The OLS slope estimator minimizes the sum of the squared raw residuals:

$$\hat{\beta}_{\text{OLS}} := \arg \min_{\beta} \sum_{i=1}^n r_i^2 = \dots \stackrel{\text{linear}}{=} (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

* **Weighted least squares (WLS)**

When errors are conditionally heteroscedastic, i.e., have different variance values σ_i , OLS won't be efficient among linear unbiased estimators. Instead, the efficient linear unbiased estimator is WLS, which requires to know these $\{\sigma_i\}$ and minimizes a *weighted* sum of squared disturbances, using weights $w_i \propto \frac{1}{\sigma_i^2}$:

$$\hat{\beta}_{\text{WLS}} := \arg \min_{\beta} \sum_{i=1}^n w_i r_i^2 = \dots \stackrel{\text{linear}}{=} (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y}$$

* **Generalized least squares (GLS)**

When errors are conditionally heteroscedastic or correlated, i.e., when the error covariance matrix $\Omega := \text{cov}[\mathbf{u}|\mathbf{X}] \neq \sigma^2 \mathbf{I}_n$, OLS will be inefficient among linear unbiased estimators. Instead, the efficient linear unbiased estimator is GLS, which requires to know Ω and minimizes the squared *Mahalanobis length*¹³ of the candidate discrepancy vector. Ω is usually unknown; when replaced by an estimator $\hat{\Omega}$, the resulting procedure is feasible GLS.

$$\hat{\beta}_{\text{GLS}} := \arg \min_{\beta} \mathbf{r}'\Omega^{-1}\mathbf{r} = \dots \stackrel{\text{linear}}{=} (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{Y}, \quad \hat{\beta}_{\text{FGLS}} := \arg \min_{\beta} \mathbf{r}'\hat{\Omega}^{-1}\mathbf{r}$$

• **Least (symmetric) absolute error**

We are interested in minimizing a different measure of loss: the absolute loss, $L(e) = |e|$, in order to have an estimator that is more robust to outliers. The optimal fit results in the **conditional median** $m(\mathbf{X}; \beta_{\text{LSA}}) := \arg \min_{m(\cdot)} \mathbb{E}[|Y - m(\mathbf{X}; \beta)|] = \dots = \text{med}_{Y|\mathbf{X}}$.

• **Least asymmetric absolute error**

We can generalize to an asymmetric loss function: $L_{\alpha}(e) := \begin{cases} (1-\alpha)|e| & \text{if } e < 0 \\ \alpha|e| & \text{if } e \geq 0 \end{cases} = (\alpha - \mathbb{1}\{e < 0\}) \times e$, which places a different penalty on overprediction and underprediction. The optimal fit is the **conditional quantile** $m(\mathbf{X}; \beta_{\text{LAA}, \alpha}) := \arg \min_{m(\cdot)} \mathbb{E}[L_{\alpha}(Y - m(\mathbf{X}; \beta))] = \dots = Q_{Y|\mathbf{X}}(\alpha)$.

Note: We can equivalently phrase all of the above in terms of the objective of minimizing loss, or of making predictions about a specific aspect of the outcome distribution:

Objective: minimize loss	Objective: prediction	Optimal estimator
$\min L(\mathbf{u}) := \mathbf{u}'\mathbf{W}\mathbf{u}, \mathbf{W} > 0$ ¹⁴	predict $\mathbb{E}_{Y \mathbf{X}}$	$\hat{\beta}_{\text{LS}} := \arg \min_{\beta} (\mathbf{Y} - m(\mathbf{X}; \beta))' \mathbf{W} (\mathbf{Y} - m(\mathbf{X}; \beta))$
$\min L(\mathbf{u}) := \mathbf{u} $	predict $\text{med}_{Y \mathbf{X}}$	$\hat{\beta}_{\text{LSA}} := \arg \min_{\beta} \sum_i Y_i - m(\mathbf{X}_i; \beta) $
$\min L_{\alpha}(\mathbf{u}) := (\alpha - \mathbb{1}\{u < 0\})u$	predict $Q_{Y \mathbf{X}}(\alpha)$	$\hat{\beta}_{\text{LAA}, \alpha} := \arg \min_{\beta} \sum_i L_{\alpha}(Y_i - m(\mathbf{X}_i; \beta))$

3.2.2 Estimator properties

Multiple estimators are often available to estimate some summary of the relationship between Y and X . How one chooses between them (besides their ease of computation) is motivated by their **statistical properties**.

¹³The Mahalanobis distance is a measure of the distance between a point A and a distribution $\mathcal{D}$. It is a multi-dimensional generalization of the idea of measuring how many standard deviations away A is from the mean of $\mathcal{D}$. It is unitless and scale-invariant, and takes into account the correlations of the data set; it is not computed separately for each residual r_i , it is a distance for the entire vector $\mathbf{r}$.

¹⁴Where $\mathbf{W}$ is a specific weighting matrix: $\mathbf{I}_n$ in OLS, $\text{diag}(w_i, \propto 1/\sigma_i^2)$ in WLS, Ω^{-1} in GLS, and $\hat{\Omega}^{-1}$ in FGLS.

Let $\hat{\theta}$ be an estimator for the population parameter θ , for a sample of size n . $\hat{\theta}$, as a function of the random sample, is a random variable. The various possible samples of size n would each lead to a different realization $\hat{\theta}^{\text{obs}}$, which together form the estimator's density or probability distribution function (pdf) $\pi_{\hat{\theta}}$. $\hat{\theta}$ has:

- finite sample properties: characteristics of $\pi_{\hat{\theta}}$ for a finite n . *Ex: bias, efficiency (precision);*
- asymptotic properties: characteristics of $\pi_{\hat{\theta}}$ as $n \rightarrow \infty$. *Ex: consistency, asymptotic distribution.*

As we always deal with finite samples, finite sample properties may seem the most important. In effect, bias (concerned with the center of the pdf) and efficiency (its spread) are the most common selection criteria. We note however that these tell us nothing about the properties of the estimator *for our own sample*, they tell us only about the distribution of values from hypothetical samples.

Finite sample properties

- $\hat{\theta}$ is **unbiased** iff $\mathbb{E}[\hat{\theta}] = \theta$

The estimator is correct *in expectation* over all possible samples. I.e., its distribution is centered around the estimand. Δ But our estimate from our own sample could be anywhere within that distribution, e.g., far from its center. The bias of $\hat{\theta}$ is $\mathbb{E}[\hat{\theta}] - \theta$.

- $\hat{\theta}$ is **efficient** or “best” iff it has the lowest possible variance of all estimators: $\mathbb{V}[\hat{\theta}] \leq \mathbb{V}[\tilde{\theta} \dots]$
Its distribution is condensed, thus though the realization from any given sample could be anywhere within that distribution, it will never be too far away from the mean (which, if the estimator is unbiased, is the true θ). For unbiased estimators, that variance is the Cramér-Rao lower bound.

Asymptotic properties

- $\hat{\theta}$ is **asymptotically unbiased** iff $\mathbb{E}[\hat{\theta}] \xrightarrow[n \rightarrow +\infty]{p} \theta$

- $\hat{\theta}$ is **asymptotically efficient** iff $\mathbb{V}[\hat{\theta}] \xrightarrow[n \rightarrow +\infty]{p} \text{asymptotic Cramér-Rao lower bound}$

- $\hat{\theta}$ is **consistent** iff $\hat{\theta} \xrightarrow[n \rightarrow +\infty]{p} \theta$

I.e., as we get enough data, then we know the truth. Sufficient conditions are that (i) $\hat{\theta}$ be asymptotically unbiased, and (ii) its variance $\xrightarrow[n \rightarrow +\infty]{} 0$.

See for example in Appendix A.1 the properties of the OLS and ML estimators in the case of the classical linear regression model.

3.2.3 Uncertainty about θ : computing $\text{SE}[\hat{\theta}]$'s and $\text{CI}_{95\%}[\hat{\theta}]$'s

i. The uncertainty in any estimand can be expressed by a $\text{CI}_{95\%}$

We are interested in a population parameter with unknown true value θ . We have identified an estimator $\hat{\theta}$ with $n - p$ degrees of freedom for that estimand, and have computed its realization $\hat{\theta}^{\text{obs}}$ for our sample. As different samples would have led to different realizations of $\hat{\theta}$, $\hat{\theta}$ has a sampling probability distribution. If the distribution is rather condensed, i.e., the standard deviation is low *relative to the estimate*, it means that $\hat{\theta}$ has high sampling precision. We would want to quantify this by computing $\text{SD}[\hat{\theta}]$ – and then use it to construct confidence intervals and test statistics. As we don't observe the sampling distribution, we don't observe $\text{SD}[\hat{\theta}]$. However, we can estimate it, and we'll call that estimator a “standard error” $\text{SE}[\hat{\theta}]$.

For any estimand θ and corresponding $\hat{\theta}$ estimator, we define:

- A **standard error** $\text{SE}[\hat{\theta}] := \widehat{\text{SD}}[\hat{\theta}]$ = an estimator of the standard deviation of $\hat{\theta}$'s sampling distribution.
- A **95% confidence interval** $\text{CI}_{95\%}[\hat{\theta}]$ for θ constructed from $\hat{\theta}$ = a range of values s.t. “if we drew all possible random samples of size n , this interval would contain the true θ in 95% of the samples.”

Correctly interpreting the CI This confidence interval is based on a *sampling* distribution; the confidence refers to our uncertainty about the *sampling* method. The CI is therefore only correctly interpreted in terms of repeated samples. I.e., we have a 95% *confidence level* that this interval estimator contains the true value θ , with the understanding that we'll be wrong 5% of the time.¹⁵ Another suggested name is “compatibility intervals,” to indicate that they give a range of parameter values that are most compatible with our data and model/assumptions (Gelman and Greenland, 2019).

ii. Traditional approach: asymptotic theory

Consider an estimand θ and an estimator $\hat{\theta}$.

1. Standard Error $\text{SE}[\hat{\theta}]$

- Example 1: θ is the population mean μ_X , $\hat{\theta}$ is the sample mean $\bar{X}$.
 - Population: X 's mean μ_X and variance σ_X^2 are unobservable.
 - Random sample: We define as estimator the sample mean $\bar{X}$. Its standard deviation $\text{SD}[\bar{X}] = \sqrt{\text{V}[\bar{X}]} = \frac{\sigma_X}{\sqrt{n}}$ is unobservable, as the population standard deviation σ_X is unobservable. A reasonable estimator for σ_X that is observable is the *sample* standard deviation s_X . We can thereby estimate $\text{SD}[\bar{X}]$ by $\text{SE}[\bar{X}] := \frac{s_X}{\sqrt{n}}$.
 - Realized sample: We compute $\text{SE}[\bar{X}]^{\text{obs}} = \frac{s_x}{\sqrt{n}}$.
- Example 2: θ is the regression slope vector β_{OLS} in a multivariate linear regression model with p regression coefficients (including the intercept).
 - Population: parameter β and error variance σ^2 are unobservable.
 - Random sample: We define the parameter estimator $\hat{\beta} \sim (\beta, \text{V}[\hat{\beta}])$. The formula of $\text{V}[\hat{\beta}]$ is known but its value unobservable — as it is notably a function of σ . For simplicity, consider the simple case of normal errors: $u|X \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$. Then $\text{SD}[\hat{\beta}] = \sqrt{\text{V}[\hat{\beta}]} = \sqrt{\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}}$. We can consistently estimate the population variance σ^2 by the bias-adjusted *sample* variance $s^2 := \frac{1}{n-p} \sum_i u_i^2$.¹⁶ We can thereby estimate $\text{SD}[\hat{\beta}]$ by $\text{SE}[\hat{\beta}] := \sqrt{\frac{1}{n-p} \sum_i u_i^2 (\mathbf{X}'\mathbf{X})^{-1}}$.
 - Realized sample: We compute $\text{SE}[\hat{\beta}]^{\text{obs}} = \sqrt{\frac{1}{n-p} \sum_i r_i^2 (\mathbf{X}'\mathbf{X})^{-1}}$.

2. Confidence Interval $\text{CI}_{95\%}[\hat{\theta}]$

We want to give a range of compatible estimates for θ . Consider *the estimator* $\frac{\hat{\theta}-\theta}{\text{SE}[\hat{\theta}]}$ of the centered and standardized estimator $\frac{\hat{\theta}-\theta}{\text{SD}[\hat{\theta}]}$.

- Example 1: $\hat{\theta} := \bar{X}$
 $\frac{\bar{X}-\mu}{\sqrt{s^2/n}}$ has a Student's $\mathcal{T}$ distribution with $n-1$ degrees of freedom. Hence, by definition:

$$\begin{aligned} & \Pr\left(q_{\mathcal{T}_{n-1}}(0.025) \leq \frac{\bar{X}-\mu}{\sqrt{s^2/n}} \leq q_{\mathcal{T}_{n-1}}(0.975)\right) = 0.95 \\ \iff & \Pr\left(\bar{X} - q_{\mathcal{T}_{n-1}}(0.975) \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} - q_{\mathcal{T}_{n-1}}(0.025) \frac{s}{\sqrt{n}}\right) = 0.95 \end{aligned}$$

where $q_{\mathcal{T}_{n-1}}()$ is the quantile function of the $\mathcal{T}_{n-1}$ distribution.

We can thus define the 95% CI

$$\text{CI}_{95\%}[\bar{X}] := \left[\bar{X} - q_{\mathcal{T}_{n-1}}(0.975) \frac{s}{\sqrt{n}} ; \bar{X} - q_{\mathcal{T}_{n-1}}(0.025) \frac{s}{\sqrt{n}} \right] = \left[\bar{X} \pm q_{\mathcal{T}_{n-1}}(0.975) \frac{s}{\sqrt{n}} \right]$$

and compute its realized value $\text{CI}_{95\%}[\bar{X}]^{\text{obs}} = \left[\bar{x} \pm q_{\mathcal{T}_{n-1}}(0.975) \frac{s}{\sqrt{n}} \right]$.

¹⁵This probability statement applies to the interval, not to β . It says $\Pr_{\text{CI}}(\beta \in \text{CI}(Y) \mid \beta) = 95\%$. This is different from saying “there is a 95% probability that the true β lies within this range” or $\Pr_{\beta}(\beta \in \text{CI}(Y) \mid \text{CI}) = 95\%$. In the frequentist paradigm, probabilities concern repeatable events, so probability statements on a fixed value such as β do not make sense.

¹⁶The bias here refers to that from the reduced degrees of freedom stemming from estimating the sample means.

- Example 2: $\hat{\theta} := \hat{\beta}_{\text{OLS}}$

If errors are normally distributed, then the sampling distribution of $\frac{\hat{\beta} - \beta_0}{\text{SE}[\hat{\beta}]}$ is a Student's $\mathcal{T}$ distribution with $n - p$ degrees of freedom. Hence, by definition:

$$\begin{aligned} \Pr\left(q_{\mathcal{T}_{n-p}}(0.025) \leq \frac{\hat{\beta} - \beta}{\text{SE}[\hat{\beta}]} \leq q_{\mathcal{T}_{n-p}}(0.975)\right) &= 0.95 \\ \iff \Pr\left(\hat{\beta} - q_{\mathcal{T}_{n-p}}(0.975) \text{SE}[\hat{\beta}] \leq \beta \leq \hat{\beta} - q_{\mathcal{T}_{n-p}}(0.025) \text{SE}[\hat{\beta}]\right) &= 0.95 \end{aligned}$$

We can thus define the 95% CI

$$\text{CI}_{95\%}[\hat{\beta}] := [\hat{\beta} - q_{\mathcal{T}_{n-k}}(0.975) \text{SE}[\hat{\beta}] ; \hat{\beta} - q_{\mathcal{T}_{n-k}}(0.025) \text{SE}[\hat{\beta}]] = [\hat{\beta} \pm q_{\mathcal{T}_{n-k}}(0.975) \text{SE}[\hat{\beta}]]$$

and compute its realized value $\text{CI}_{95\%}[\hat{\beta}]^{\text{obs}} = [\hat{\beta}^{\text{obs}} \pm q_{\mathcal{T}_{n-k}}(0.975) \text{SE}[\hat{\beta}]^{\text{obs}}]$.

The larger its degrees of freedom, the closer a $\mathcal{T}$ distribution gets to the standard normal distribution. Therefore, in both examples, when $n - p$ is sufficiently large, we can simply use the normal distribution:¹⁷

$$\text{CI}_{95\%}[\hat{\theta}] \simeq [\hat{\theta}^{\text{obs}} \pm q_{\mathcal{N}}(0.975) \text{SE}[\hat{\theta}]^{\text{obs}}] = [\hat{\theta}^{\text{obs}} \pm 1.96 \text{SE}[\hat{\theta}]^{\text{obs}}]$$

iii. Simulation approach: Bootstrap

The approach illustrated above often relies on the assumed *asymptotic* sampling distribution of $\hat{\theta}$. That distribution rests on asymptotic theory (which usually leads to limit normal and χ_2 sampling distributions). When our sample size is small (making this asymptotic approximation incorrect), or when analytical expressions for the uncertainty of the particular statistic $\hat{\theta}$ are complicated, i.e., when conventional analytic approximations fail, we can create an alternative sampling approximation of the finite-sample distribution of interest by **bootstrap**.

The bootstrap procedure is a way to estimate the sampling distribution of the sample statistic $\hat{\theta}$, by resampling with replacement from the current sample to generate multiple “resamples” of the same size n . From each bootstrap sample we obtain one realization of θ ; suppose we perform $B = 100$ bootstrap resamples, we hence obtain a distribution: $\{\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(B)}\}$, and can estimate $\text{SD}[\hat{\theta}]$ by their realized standard deviation.

Advantages and limits:

- + It does not assume any underlying distribution of the data.
- + It can be applied to any sample statistic.
- + Bootstrap CIs are asymptotically consistent (though we can't know the true CI) and more accurate than the traditional intervals.
- Inference still relies on an appropriately drawn sample; and assumes independent resamples. Therefore with structured models, one must think carefully about the design of the resampling procedure (e.g. with clusters, one may want to use block bootstrap instead of ordinary bootstrap).
- Simple but time-consuming.

3.3 Hypothesis testing

3.3.1 Null Hypothesis Significance Testing (NHST) paradigm

A statistical test is a method of verifying a statistical hypothesis.

A statistical hypothesis is a hypothesis on the probability distribution of a **test-statistic** T computed from the data, whose probability distribution is connected to our research question.

Microeconometrics commonly abides by the null hypothesis significance testing (NHST) paradigm. Null hypothesis testing proceeds by *reductio ad absurdum*: a hypothesis is assumed valid if its counterclaim is highly implausible.

The general approach to conducting a statistical test thereby consists of the following steps:

¹⁷As $1.96 \simeq 2$, it is common to read that “statistically significant estimates are at least two standard errors from zero.”

1. Write the null hypothesis H_0 — the hypothesis to nullify.
2. Design a test statistic T that summarizes the data's deviation from what would be expected under H_0 , and that has a specific distribution under H_0 . Ex:
 - a Wald test is a test in which the test statistic is a specific formula of the estimated parameters and their estimated covariance matrix; it has an *asymptotic* χ^2 distribution under H_0 ;
 - a t -test is any test in which the test statistic has a Student's $\mathcal{T}$ distribution under H_0 ;
 - a z -test is any test in which the test statistic has an *asymptotic* normal distribution under H_0 ;
 - an F -test is any test in which the test statistic has an $\mathcal{F}$ distribution under H_0 .
3. Compute the realized value of T for our data: T^{obs} .
4. If T^{obs} falls in the tails of the distribution, i.e., is very unlikely given H_0 , we can reasonably reject H_0 .

Common null hypotheses and corresponding tests

H_0	Test statistic T	Inference
Wald test (OLS or MLE) <i>Are the k coefficients β jointly significant?</i> $H_0 : \beta_0 = 0_{k \times 1}$, for k among the $p - 1$ slope coefficients	$W := (\hat{\beta} - \beta_0)' \widehat{\text{Var}}_{\hat{\beta}}^{-1} (\hat{\beta} - \beta_0)$	IF standard regularity conditions ¹⁸ $\implies W \stackrel{a}{\sim}_{H_0} \chi_k^2$ IF iid normal errors $\implies W \stackrel{a}{\sim}_{H_0} k\mathcal{F}_{k, n-p} \stackrel{a}{\sim} \chi_k^2$
t-test (OLS); z-test (MLE) <i>Is the coefficient β_j significant?</i> $H_0 : \beta_{j,0} = 0$	$t := \frac{\hat{\beta} - \beta_0}{\text{SE}[\hat{\beta}]}$ $z := \frac{\hat{\beta} - \beta_0}{\text{ASE}[\hat{\beta}]}$	IF standard regularity conditions $\implies t, z \stackrel{a}{\sim}_{H_0} \mathcal{N}(0, 1)$ ¹⁹ IF iid normal errors $\implies t \stackrel{a}{\sim}_{H_0} \mathcal{T}_{n-p} \stackrel{a}{\sim} \mathcal{N}(0, 1)$
F-test (OLS); Likelihood ratio-test (MLE) <i>Are the $p - 1$ slope coefficients jointly significant, i.e., is the model overall significant?</i> <i>(Is it a significantly better fit than a model with no predictors?)</i> $H_0 : \beta_0 = 0_{(p-1) \times 1}$, for all $p - 1$ slope coefficients	$F := \frac{\text{ESS}/(p - 1)}{\text{RSS}/(n - p)}$ $G^2 := D_{\text{null}} - D = 2(l - l_{\text{null}})$	IF standard regularity conditions $\implies G^2 \stackrel{a}{\sim}_{H_0} \chi_{df[D_{\text{null}} - D]}^2$ IF iid normal errors $\implies F \stackrel{a}{\sim}_{H_0} \mathcal{F}_{p-1, n-p}$

Example In the case of the OLS estimation of a linear model with p coefficients, we typically want to test the hypothesis of a relationship between Y and X_j , i.e., that $\beta_j \neq 0$.

¹⁸ Δ “Standard regularity conditions” refer to different assumptions in OLS vs MLE.

¹⁹ The asymptotic standard normal distributions of the t - and z -tests stem from the slope coefficients having asymptotic normal distributions themselves (a result of a central limit theorem). So *for large n only*, we can say that the test statistics are approximately normally distributed. However, for small n , they are close to normal only if the data themselves are normal. E.g., if n is small and the data are substantially non-normal or in the presence of outliers, the t -test can give misleading results.

1. Define H_0 We define the null hypothesis $H_0: \beta_{j,0} = 0$
2. Design T We define the t -statistic $t_{\hat{\beta}_j} := \frac{\hat{\beta}_j - \beta_{j,0}}{\widehat{SD}[\hat{\beta}_j]} = \frac{\hat{\beta}_j - 0}{SE[\hat{\beta}_j]}$. It has a known distribution under H_0 : $t_{\hat{\beta}_j} \stackrel{a}{\sim}_{H_0} \mathcal{N}(0, 1)$, and if $\hat{\beta}_j$ is normally distributed, then $t_{\hat{\beta}_j} \sim_{H_0} \mathcal{T}_{n-p}$.
3. Compute T^{obs} $t_{\hat{\beta}_j}^{\text{obs}} := t(\text{observed data}) = \frac{\hat{\beta}_j^{\text{obs}} - 0}{SE[\hat{\beta}_j]^{\text{obs}}}$
4. Interpret We define the *two-tailed*²⁰ p -value $_{\hat{\beta}_j} := \Pr\left(\text{observing a } |t| \geq |t_{\hat{\beta}_j}^{\text{obs}}| \mid H_0\right)$, i.e., the probability of observing data as extreme as that actually observed, assuming H_0 .²¹

$p\text{-value}_{\hat{\beta}_j}$ small $\iff t_{\hat{\beta}_j}^{\text{obs}}$ falls in the tail of the Student's $\mathcal{T}$ -distribution
 $\iff$ observing our $t_{\hat{\beta}_j}^{\text{obs}}$ under H_0 is highly unlikely
 $\implies$ reject H_0
 $\implies$ there is a relationship between Y and X_j ; $\hat{\beta}_j$ is “significant”

In econometrics, the standard approach is to dichotomize the evidence using a p -value threshold, usually the *significance level* $\alpha = 5\%$. $\hat{\beta}_j$ is “statistically significant” iff $p_{\hat{\beta}_j} \leq 0.05$, i.e., iff there is a less than 5% chance of observing the effect size that we observe if there was in fact no effect.

In the typical regression outputs from statistical software packages, summaries include multiple test statistics, that test notably the significance of individual slope coefficients (e.g., through t -tests as the above in OLS and z -tests in MLE) and the significance of the model as a whole, i.e., the joint significance of the $p - 1$ slope coefficients (through F -tests in OLS and likelihood-ratio tests in MLE); see Appendix B.

3.3.2 Type I/II errors, size and power

A test can lead to two types of mistakes:

- **Type 1 error** or *false positive*: $\{- \mid H_0\}$ reject H_0 when shouldn't... (*overconfident*)
- **Type 2 error** or *false negative*: $\{+ \mid H_0\}$ don't reject H_0 when should (*overcautious*)

We define a test's:

- **size** α_T = probability of erroneously rejecting H_0 $:= \Pr(\text{type 1 error}) = \Pr(- \mid H_0)$
- **power** κ_T = probability of correctly rejecting H_0 $:= 1 - \Pr(\text{type 2 error}) = \Pr(- \mid H_0)$

Intuitively, we would like to both minimize the size and maximize the power of our test. To guarantee $\alpha_T \leq 0.05$, we simply set the significance level $\alpha = 0.05$. To guarantee $\kappa_T \geq 0.80$, we need a sufficiently large sample size n , otherwise the “Minimum Detectable Effect” will be very high.²²

²⁰We can actually use the test statistic T to carry out two different tests:

- A two-tailed test: if we want to test for the possibility of the relationship in both directions. $H_0: \beta_j = 0, H_a: \beta_j \neq 0$. Both tails of T 's distribution constitute therefore the “critical region”, each containing $\frac{\alpha}{2}$ of the values. By default, statistical packages report the two-tailed p -values.
- A one-tailed test: to test for the possibility of the relationship only in one direction. E.g.: $H_0: \beta_j = 0, H_a: \beta_j > 0$. Only one tail of T 's distribution makes the critical region, containing α of the values. Only z - and t -tests can accommodate one-tailed tests. F -tests, χ^2 -tests... cannot as their distributions are not symmetric.

²¹⚠ The p -value is often misinterpreted to be the probability of the null hypothesis, whereas it is the probability of data, given the null. $p\text{-value} = \Pr(\text{obs} \mid \text{hyp}) \neq \Pr(\text{hyp} \mid \text{obs})$. I.e., if we want to make inferences about the actual values of parameters, p -values and frequentist regression fail us: p -values make inferences about the probability of data, not of parameter values. Only Bayesian methods allow us to make inferences about the actual values of the parameters, e.g., to assess the probability of the null. To summarize: a frequentist's conclusion (in the form of a p -value) is a statement considering that data is random and model parameters are fixed, whereas a Bayesian's conclusion (in the form of a credible interval) is a distribution of parameter values that would generate the fixed data.

²²In hypothesis testing in econometrics, we typically want at least 80% power and a maximum size of 5%. I.e., we accept to incorrectly reject the null a maximum of 5% of the time, and to correctly reject it at least 80% of the time (i.e., 80% of studies conducted with a given sample size will correctly reject the null). $95\% > 80\%$: econometrics is more focused on avoiding

Power calculations Having adequate power means that if there really is an effect, the empirical strategy and data will enable the test to detect it. Low-powered studies will instead *miss* the effect.²³ Post-estimation, it is useful to perform a retrospective design analysis and ask: “*Was my study sufficiently powered?*”, especially if we found a statistically significant non-null effect. But it must be done correctly:

⚠ To estimate power one must first postulate a ‘true’ effect size, which can be thought of as that observed in an infinitely large sample. That effect size should be determined from a literature review, not the effect size observed in one’s study! The latter is noisy, and generally overestimated (publication bias), and would therefore lead to overestimates of power.

3.3.3 Criticisms of the NHST and ‘statistical significance’

The binary approach to statistical hypothesis testing, based on the NHST falsificationist paradigm (where the underlying truth is H_0 “no effect” or H_a “effect”) where the outcome is a binary statement of ‘statistical significance’ from a p-value threshold, is heavily criticized. It is argued that:

- The underlying reality is not a simple Yes or No: in social sciences, the null hypothesis of zero effect is generally implausible — there are virtually no true zeros — and thus uninteresting. The null model is very false, so we are very likely to reject it with enough data.
 - ↔ Instead, we need to find alternatives to thinking in terms of conditional independence in order to study causality. One idea would be to estimate these dependences directly, rather than modeling the world in terms of conditional independence and estimating this structure through tests of null effects.
- Interpreting p-values dichotomously loses a lot of information.
 - ↔ Instead, one could interpret p-values continuously, the strength of evidence for H_0 being a continuous function of the p-value. A low signal-to-noise ratio implies high uncertainty in our conclusions, which is fine. Even in the context of policy-relevant research, a policy can still be recommended in the midst of such uncertainty. The goal of a statistical analysis should be to learn what we can from the data and model and to get a sense of what we don’t know, not to force an expression of near-certainty from data.
- Interpreting p-values dichotomously may induce selection bias for overestimates.
 - If to be publishable, estimates must be ‘significant’, i.e., more than two standard errors away from 0; this will select for overestimates. Specifically, statistically significant results are often subject to type M (magnitude) errors—i.e., being overestimated— and even type S (sign) errors—being in the wrong direction (Gelman et al., 2020, p.59).
 - Especially in the social sciences, “a world in which measurements are noisy and effects are small”, i.e., where studies are likely to be underpowered, then selection on statistical significance is particularly harmful, leading Gelman (2018) to argue that “*the current replication crisis in science arises in part from the ill effects of null hypothesis significance testing being used to study small effects with noisy data.*” Ioannidis et al. (2017)’s large survey of the empirical economics literature finds in fact that the median statistical power is only 18%, and nearly 80% of the reported effects are exaggerated, typically by a factor of two.

overconfidence than worried about being overcautious. Note also that having a high sample size n is not sufficient to have higher statistical power — empirical studies have actually found zero or weak correlations between the two. The power of a study depends indeed on the sample size, the true size of the effect, measurement variance, and the number of comparisons performed. Note finally that studies of small effects, although potentially important, are unlikely to be statistically significant because they have insufficient power to detect the magnitudes of effects.

²³Lacasse et al. (2020) is a good example of this. Rephrasing their specific independent and dependent variables as generic X and Y : “*Because enrollment in the trial was stopped before we had reached our proposed sample size, the trial was underpowered, with the consequence of a wide confidence interval around the point estimate. [...] The data that were accrued could not rule out benefit or harm from X .*” As summarized in the abstract: “*Our underpowered trial provides no indication that X has a positive or negative effect on Y .*”

4 Statistical inference [under a Bayesian approach]

One of the main distinguishing features of Bayesian inference is the expression of all information, including uncertainty, using probability. From this paradigm stem new possibilities at every step of inference.

4.1 Steps of Bayesian inference

We start with the same inferential situation: a population parameter has an unknown true value θ , and we want to learn about it using data.

- (0) Definition of uncertainty about θ : While in frequentist inference θ is considered fixed, Bayesian inference represents the uncertainty about it by using a *random variable* Θ . At all times, Θ has a probability distribution which represents our belief or uncertainty about its actual value.
- (1) Prior: Prior to observing data, Θ has a *prior* distribution that represents our belief about it.
- (2) Bayes' rule: After observing the data, we use Bayes' theorem to combine the prior distribution with the information contained in the data to 'update our belief about Θ ' and produce a *posterior* distribution:

$$\text{posterior} \rightarrow \Pr(\theta | \text{data}) = \frac{\Pr(\theta, \text{data})}{\Pr(\text{data})} = \frac{\Pr(\theta) \Pr(\text{data} | \theta)}{\Pr(\text{data})} \propto \text{prior} \Pr(\theta) \Pr(\text{data} | \theta) \leftarrow \text{likelihood}^{24}$$

In the canonical regression setting, with α a hyperparameter of Θ 's distribution, Bayes' theorem gives²⁵

$$\pi_{\Theta|Y,X}^{\text{post}}(\theta | y, X; \alpha) = \frac{\pi_{\Theta}^{\text{pre}}(\theta; \alpha) f_{Y|X,\Theta}(y | X, \theta)}{m_{Y|X}(y | X; \alpha)} \propto \pi_{\Theta}^{\text{prior}}(\theta; \alpha) f_{Y|X,\Theta}(y | X, \theta)$$

- (3) DGP inference: We summarize posterior uncertainty about Θ by reporting quantities such as:
 - a measurement of central tendency, e.g., the posterior mean $\mathbb{E}[\Theta|Y,X]$ or median $\text{med}[\Theta|Y,X]$;
 - a 95% *credible* interval $C_{95\%}$ s.t. “given the observed data, there is a 95% probability that the true value of Θ lies within this range”:

$$\Pr(\Theta \in C_{95\%} | y, X) = \Pi_{\Theta|Y,X}^{\text{post}}(C_{95\%} | y, X) = 0.95$$

If the goal is prediction rather than DGP inference, we proceed similarly up to the last step, and then form a posterior predictive distribution:

- (3') Prediction: Let $\tilde{Y}$ and $\tilde{X}$ denote a generic future random outcome and its covariates. We propagate the uncertainty about Θ into predictions of $\tilde{Y}$ (analytically if the posterior and the likelihood enable it by having known closed forms, by simulations otherwise).

- Before observing the data, the *prior* predictive distribution of $\tilde{Y}$ is²⁶

$$p_{\tilde{Y}|\tilde{X}}^{\text{prior}}(\tilde{y} | \tilde{x}; \alpha) = \int p_{\tilde{Y}|\tilde{X},\Theta}(\tilde{y} | \tilde{x}, \theta) \pi_{\Theta}^{\text{prior}}(\theta; \alpha) d\theta$$

- After observing the data, the *posterior* predictive distribution of $\tilde{Y}$ is

$$\begin{aligned} p_{\tilde{Y}|\tilde{X},Y,X}^{\text{post}}(\tilde{y} | \tilde{x}, y, X; \alpha) &= \int p_{\tilde{Y}|\tilde{X},Y,X,\Theta}(\tilde{y} | \tilde{x}, y, X, \theta) \pi_{\Theta|Y,X}^{\text{post}}(\theta | y, X; \alpha) d\theta \\ &=^{27} \int p_{\tilde{Y}|\tilde{X},\Theta}(\tilde{y} | \tilde{x}, \theta) \pi_{\Theta|Y,X}^{\text{post}}(\theta | y, X; \alpha) d\theta \end{aligned}$$

²⁴The difference with the frequentist likelihood is that it is now also conditioned on Θ . Note: For simplicity, in this formula we have used the probability operator. However, formally, the objects are the probability distributions or densities.

²⁵The denominator is the prior predictive density of the observed data, or marginal likelihood: $m_{Y|X}(y|X; \alpha) := \int p_{Y|X,\Theta}(y|X; \theta) \pi_{\Theta}^{\text{prior}}(\theta; \alpha) d\theta$. It is sometimes called the “evidence”. In effect, it is a scaling factor.

²⁶Where we assume that Y and $\tilde{Y}$ follow the same conditional distributional rule, hence $p_{\tilde{Y}|\tilde{X},\Theta}(\cdot | \cdot) = p_{Y|X,\Theta}(\cdot | \cdot)$.

²⁷Under the usual assumption that the future outcome and the observed outcomes are conditionally independent given the parameter and their covariates $\tilde{Y} \perp Y | \Theta, \tilde{X}, X$.

Advantages/Distinctions of Bayesian inference w.r.t. frequentist inference

👉 Intuitive interpretation of findings

Because uncertainty is encoded probabilistically, our uncertainty about θ after observing the data is represented by a distribution of values. We can effortlessly compute a 95% credible interval from the posterior, with an intuitive interpretation: “*There is a 95% chance that this CI contains the true θ .*” In comparison, the frequentist 95% confidence interval refers to our uncertainty about the *sampling method*—not θ —and is interpreted in terms of repeated samples: “*If we drew all possible random samples of size n , this interval would contain the true θ in 95% of the samples*” (see Section 3.2.3). More generally, the Bayesian framework enables us to actually answer questions like “*What is the probability that $\Theta = 4$?*”, while the frequentist framework provides more convoluted estimates: the probability of the data if $\theta = 4$.

👉 Including prior information

Bayesian inference provides a framework for combining multiple sources of information, namely, prior information and data.

In the scalar Gaussian-Gaussian conjugate model, the posterior mean is actually a weighted average of the prior mean and the data-based estimator, with each weight proportional to the inverse of the corresponding variance (i.e., a precision-weighted average). And as a result, the posterior standard deviation $SD_{\theta|Y,X}^{\text{post}}$ is strictly smaller than both the prior standard deviation $SD_{\Theta}^{\text{prior}}$ and the sampling standard error of the data-based estimator $SD_{\hat{\theta}|\Theta}^{\text{sampling}}$.

👉 Making predictions is facilitated by the computation often being ~~optimization~~-simulation-based

A Bayesian will argue that one should do predictions based on the whole posterior distribution of possible coefficient values, rather than basing prediction on a point estimate and disregard all information about how imprecise this estimate is. Because the uncertainty about θ is encoded in a probability distribution, we can propagate this uncertainty into predictions of a new random outcome $\tilde{Y}$, by using simulations. We draw a value from Θ 's posterior distribution and make a *probabilistic* prediction of a new outcome for this value of Θ , and repeat this simulation S times. The S resulting values make the *posterior predictive distribution*.

4.2 Choosing Θ 's prior distribution

We include additional information about Θ through our choice of a prior distribution. **Considerations to guide that choice:**

- Uninformative (flat) vs weakly informative priors:
 - Using an uninformative or “flat” prior (the uniform distribution) results in the posterior distribution being equal to the product of the likelihood and a mere constant, s.t. the mode of the posterior distribution is the ML or LS estimator.
 - Weakly informative priors: “*What you should be doing when you think you want to use noninformative priors*” (https://statmodeling.stat.columbia.edu/2009/05/24/handy_statistic/). Ex: The R function `rstanarm::stan_glm()` adjusts the default priors based on the scale of the variables in the model.
- “Conjugate” prior probability distributions (for the ... distribution): the posterior distributions $f(\theta|x)$ are in the same family as the prior probability distribution $f(\theta)$.

4.3 Estimating Θ 's posterior distribution

When the prior and likelihood combine into a posterior distribution with a known closed form, we can derive and study this posterior analytically. Often, however, the posterior as defined by Bayes' rule does not have a distribution that we can manipulate or sample from directly. In that case, we can use simulations to form an empirical distribution, which we'll use as an approximation of the posterior.

Specifically, we use that $\pi^{\text{post}}(\theta|Y, X) \propto f(Y|X, \theta)\pi^{\text{prior}}(\theta)$ to construct a numerical sampling procedure whose output approximates posterior draws. Informally, the prior and likelihood are sufficient to derive some relative

posterior plausibility of candidate values.²⁸ We thereby generate a set $\{\tilde{\theta}^s\}_{s=1}^S$, i.e., an empirical distribution, which forms our approximation of the posterior $\Pi_{\Theta|Y,X}^{\text{post}}$.

4.4 Propagating posterior uncertainty

We can use simulations to make inferences about quantities or distributions of interest beyond the posterior. Let us omit covariates in the formulas below for simplicity. Starting from S posterior draws $\{\tilde{\theta}^s\}_{s=1}^S$, we use each draw to compute another random quantity. The resulting empirical distribution then approximates the distribution of the object of interest as $S \rightarrow \infty$. For example:

- To approximate the posterior distribution of a function $g(\Theta)$:

$$\left\{ \begin{array}{l} \tilde{\theta}^1 \sim \Pi_{\Theta|Y}^{\text{post}}(\cdot | y) \rightarrow \text{compute } g(\tilde{\theta}^1) \\ \dots \\ \tilde{\theta}^S \sim \Pi_{\Theta|Y}^{\text{post}}(\cdot | y) \rightarrow \text{compute } g(\tilde{\theta}^S) \end{array} \right. ; \text{ The empirical distrib. of } \left\{ \begin{array}{l} g(\tilde{\theta}^1) \\ \dots \\ g(\tilde{\theta}^S) \end{array} \right\} \xrightarrow{S \rightarrow \infty} \mathcal{D}_{(g(\Theta)|Y)}(\cdot | y)$$

- To approximate the posterior predictive distribution of $\tilde{Y}$:

$$\left\{ \begin{array}{l} \tilde{\theta}^1 \sim \Pi_{\Theta|Y}^{\text{post}}(\cdot | y) \rightarrow \text{compute } \tilde{y}^1 \sim f_{\tilde{Y}|\Theta}(\cdot | \tilde{\theta}^1) \\ \dots \\ \tilde{\theta}^S \sim \Pi_{\Theta|Y}^{\text{post}}(\cdot | y) \rightarrow \text{compute } \tilde{y}^S \sim f_{\tilde{Y}|\Theta}(\cdot | \tilde{\theta}^S) \end{array} \right. ; \text{ The empirical distrib. of } \left\{ \begin{array}{l} \tilde{y}^1 \\ \dots \\ \tilde{y}^S \end{array} \right\} \xrightarrow{S \rightarrow \infty} f_{\tilde{Y}|Y}(\cdot | y)$$

- To approximate $\Pr(\beta_A > \beta_B | Y=y)$

E.g., in a three-arm experiment, to evaluate the posterior probability that treatment A is more effective than treatment B , with the regression model $Y_i = \alpha + \beta_A D_{Ai} + \beta_B D_{Bi} + u_i$, where β_A and β_B are the treatment effects relative to control.

$$\left\{ \begin{array}{l} (\tilde{\beta}_A^1, \tilde{\beta}_B^1) \sim \Pi_{(\beta_A, \beta_B)|Y}^{\text{post}}(\cdot, \cdot | y) \rightarrow \text{compute } \mathbb{1}\{\tilde{\beta}_A^1 > \tilde{\beta}_B^1\} \\ \dots \\ (\tilde{\beta}_A^S, \tilde{\beta}_B^S) \sim \Pi_{(\beta_A, \beta_B)|Y}^{\text{post}}(\cdot, \cdot | y) \rightarrow \text{compute } \mathbb{1}\{\tilde{\beta}_A^S > \tilde{\beta}_B^S\} \end{array} \right. ; \frac{1}{S} \sum_s \mathbb{1}\{\tilde{\beta}_A^s > \tilde{\beta}_B^s\} \xrightarrow{S \rightarrow \infty} \Pr(\beta_A > \beta_B | y)$$

²⁸For example, the Stan language uses Markov Chain Monte-Carlo (MCMC) algorithms, which are iterative sampling algorithms. “*Monte-Carlo*” refers to the practice of estimating the properties of a distribution by examining random samples from the distribution (ex: instead of finding the mean of a normal distribution by directly calculating it from its closed form, one draws a random sample from the normal distribution and calculates the sample mean). “*Chain*” refers to the random samples being generated by a special sequential process: each random sample is used as a stepping stone to generate the next random sample. Note that this means that the draws are not independent. The “*Markov*” property of the chain indicates that each new sample depends only on the one immediately preceding it, and not on any samples before that one.

5 Model comparison

Learning from data generally has one of two ultimate objectives: DGP inference or prediction. Model comparison should proceed in line with the objective. After a brief paragraph on *nested* model discrimination, this section focuses on model comparison for prediction, our objective will therefore be predictive performance.²⁹ Much of this section is taken from [Gelman et al. \(2014\)](#).

5.1 Comparing nested models: F tests

If two models are *nested*, i.e., one represents a special case of the other, we can easily discriminate between them using a standard hypothesis test of the parametric restrictions on the nested one.

The key questions are: (1) Is the improvement in fit large enough to justify the additional difficulty in fitting, and in a Bayesian context: (2) Is the prior distribution on the additional parameters reasonable?

5.2 Comparing non-nested models: IC, CV

We want to determine which model best predicts new data generated from the true—and unknown—DGP P_0 . Ideally, we would measure each model's out-of-sample predictive performance on such new data.

Consider an out-of-sample set of random variables $\tilde{Y} = (\tilde{Y}_1, \dots, \tilde{Y}_m)' \sim P_0$, with realizations $\tilde{y} = (\tilde{y}_1, \dots, \tilde{y}_m)'$. In practice, we extract such a test set $\tilde{Y}$ (and keep the rest of the data as training set Y) before fitting the models.

A candidate model M 's predictive performance can be defined in different ways:

- When predicting a *point* value, predictive accuracy can be evaluated using a loss function, such as absolute-error loss or squared-error loss. Using the test set, we aggregate the prediction errors into a summary measure, such as the Mean Absolute Error (MAE) or the Root Mean Squared Error (RMSE):³⁰

$$\text{MAE} := \frac{1}{m} \sum_i |\hat{Y}_i - \tilde{Y}_i|, \quad \text{RMSE} := \sqrt{\text{MSE}} := \sqrt{\frac{1}{m} \sum_i (\hat{Y}_i - \tilde{Y}_i)^2}$$

- A more general measure evaluates the model's entire predictive distribution, by estimating its **expected log predictive density (ELPD)**.

Let $p_{\tilde{Y}|Y}^M(\cdot | Y)$ denote the model's fitted predictive density given the training data. Its ELPD is

$$\text{ELPD}^M(Y) := \mathbb{E}_{\tilde{Y} \sim P_0} [\log p_{\tilde{Y}|Y}^M(\tilde{Y} | Y)]$$

If the fitted model is summarized by a point estimator $\hat{\theta}^M(Y)$	If the fitted model is summarized by a posterior distribution $\pi_{\Theta Y}^{M,\text{post}}(\cdot Y)$
▷ Fitted predictive density for a new realization $\tilde{y}$: $p^{M,\text{plug}}(\tilde{y} Y) := p^M(\tilde{y} \hat{\theta}^M)$ ▷ Expected log predictive density: $\text{ELPD}^{M,\text{plug}}(Y) = \mathbb{E}_{\tilde{Y} \sim P_0} [\log p^{M,\text{plug}}(\tilde{Y} Y)]$	$p^{M,\text{post}}(\tilde{y} Y) := \int p^M(\tilde{y} \theta^M) \pi^{M,\text{post}}(\theta^M Y) d\theta^M$ $\text{ELPD}^{M,\text{post}}(Y) = \mathbb{E}_{\tilde{Y} \sim P_0} [\log p^{M,\text{post}}(\tilde{Y} Y)]$

In practice, P_0 is unknown and new observations have not yet been observed, so the ELPD cannot be computed directly from the available data. We can estimate it—the estimator hence remains subject to sampling variability. A natural estimator of the expected LPD for *new* data is the LPD for *existing* data, but it will be overly optimistic. We hence use estimators which estimate some LPD for observed data and adjust it by subtracting an approximate bias correction:

²⁹In classical econometrics focused on DGP inference, especially when the goal is causal inference, the research design drives the model specification such that there isn't much foray into model comparison and selection.

³⁰Note that both measures have the same unit as Y , which makes them easier to interpret than e.g. the MSE, but it means that they are scale-dependent, and hence can only be compared across models in the same units. The RMSE is the standard deviation of the residuals, i.e., of the unexplained variation. Compared to the MAE, the RMSE penalizes large errors more.

– Information Criteria (IC)

We compute scores that start from the LPD, and subtract from it the number of estimated parameters to penalize model complexity, i.e., to not overfit the data. They are generally expressed on the deviance scale; the model with smallest IC is preferred.³¹

- * The **Akaike information criterion (AIC)** uses as starting point for the LPD a point estimator: $\hat{\theta}^M$, and applies an *asymptotic* correction by subtracting the number of estimated parameters k :³²

$$\text{AIC} := -2 \left(\widehat{\text{ELPD}}_{\text{AIC}} \right) := -2 \left(\widehat{\text{LPD}}_{\text{AIC}} - k \right) = -2 \log p^M(\tilde{Y} | \hat{\theta}^M) + 2k$$

The AIC corrected for small samples is $\text{AIC}_c = \text{AIC} + \frac{2k(k+1)}{n-k-1} \xrightarrow{n \rightarrow +\infty} \text{AIC}$

- * The **Watanabe-Akaike information criterion (WAIC)** uses as starting point for the LPD the whole posterior distribution $\pi_{\Theta|Y}^{M, \text{post}}(\cdot | Y)$, and adjusts for overfitting by using a Bayesian *effective* number of parameters based on the posterior distribution.

– K-fold cross-validation

We repeatedly partition the data into a training set and a validation set, fit the model to the training set, and evaluate its predictive performance on the held-out set. Specifically in K -fold CV the data are randomly partitioned into K equal-sized sets $\{Y_v\}_{v=1}^K$. $K = 10$ is commonly used. The CV process is repeated K times, using a different set for validation each time, such that each observation is used for validation exactly once. We then average the K resulting held-out LPDs into one estimator:

$$\widehat{\text{ELPD}}_{K\text{-CV}} := \sum_{v=1}^K \widehat{\text{LPD}}(Y_v) = \sum_{v=1}^K \log p^{M, \text{plug/post}}(Y_v | Y_{-v})^{33}$$

Notes:

- * In any CV process, each prediction is conditioned on $n - v$ data points instead of n , which causes underestimation of the predictive fit. We can correct for this bias by estimating how much better predictions would be obtained if conditioning on n data points (Gelman et al., 2014).
- * ‘**Leave-one-out**’ (LOO) CV corresponds to n -fold CV, i.e., the extreme case of n partitions, where each validation set is made of a single data point.

Which estimator to choose? Neither cross-validation nor information criteria are perfect. AIC does not work in settings with strong prior information, WAIC relies on a data partition unamenable to structured models such as for spatial or network data, cross-validation is computationally expensive as getting a stable estimate requires many data partitions and fits. Gelman et al. (2014)’s preferred choice is “cross-validation, with WAIC as a fast and computationally convenient alternative. WAIC is fully Bayesian (using the posterior distribution rather than a point estimate) [...]. A useful goal of future research would be a bridge between WAIC and cross-validation with much of the speed of the former and robustness of the latter.”

³¹IC are based on the likelihood function, and hence can only be used to choose between models estimated using maximum likelihood. E.g., they cannot be used with models fitted using pseudo maximum likelihood, or to compare models with different fixed effects that were fitted using restricted maximum likelihood (REML), as residual likelihoods are not comparable.

³²This bias correction is based on the asymptotic normal posterior distribution, for which simply subtracting k corrects for the number of parameters. Δ When we go beyond linear models with flat priors, e.g., models with hierarchical structures or informative priors, the number of *effective* parameters isn’t k so one can’t simply subtract k .

³³CV can use for the LPD any starting point, i.e., either averaging over the posterior distribution $\pi_{\Theta|Y}^{M, \text{post}}(\cdot | Y)$ or conditioning on a point estimator $\hat{\theta}^M(Y)$. In Bayesian CV using simulations, we assume that we can summarize the posterior by S simulation draws $\theta^1, \dots, \theta^S$. We can then estimate the log predictive density for Y_v as: $\widehat{\text{LPD}}(Y_v) = \ln \frac{1}{S} \sum_{s=1}^S p(Y_v | \theta^s)$, hence $\widehat{\text{ELPD}}_{K\text{-CV}} = \sum_{v=1}^K \ln \left(\frac{1}{S} \sum_{s=1}^S p(Y_v | \theta^s) \right)$.

6 Other branches of statistical modeling

6.1 Statistical inference using agent-based models

Agent-based models (ABMs) are computational models^a that simulate the actions and interactions of autonomous agents within a system, to assess their effects on the system as a whole. The goal is to re-create and predict the emergence^b of higher-level system properties from simple agent-level behaviors, via a “bottom-up” approach.

ABMs are generally composed of 3 elements:

1. many **agents** with assigned attributes;
2. simple **rules** about: their individual decision-making process, how they interact, how they learn and adapt—these rules can be deterministic or probabilistic;
3. an **environment**.

^aComputational models are mathematical models that study the behavior of a system by computer simulation. The system studied is often a complex nonlinear system for which simple analytical solutions are not available. Experimentation is therefore done by modifying the model’s parameters, and comparing outcomes. Ex: weather forecasting models, flight simulator models, neural network models, ABMs...

^bThe process of *emergence* can be expressed as “the whole is greater than the sum of its parts”.

Goal of ABMs ABMs allow us to observe how the behaviors of individual agents affect the system as a whole and if any emergent structure develops within the system. They show how small-scale changes can affect large-scale outcomes within the system.

At a formal level, an ABM is just a statistical model. But agent-based modeling differs from other types of statistical modeling in that it describes only the behavior of the agents in a system, rather than global properties of the system.

Use in different fields

- In economics: ABMs can describe the microeconomic actions of adaptive agents, which give rise to emergent behavior in the form of macroeconomic structures—which, in turn, influence agent decisions. Ex: we can represent the economy as a complex system, with crashes and booms that emerge from non-linear responses to small changes.
- In ecology: ABMs are often called individual-based models (IBMs) and are used to study population dynamics, plant-animal interactions...
- In epidemiology: epidemiological ABMs now complement traditional compartmental models (such as the deterministic SIR—Susceptible/Infectious/Recovered—model) which they have tended to surpass in terms of prediction accuracy to model the spread of epidemics.

Statistical inference

1. Model validation and selection, uncertainty quantification, and fitting ABMs to data: **There does not seem to be formal guidelines and procedures from the statistical literature, for: fitting ABMs to data, making quantified statements of uncertainty about the outputs (e.g., calculating confidence intervals on predictions), nor testing whether a specific parameter (rule) is needed in an ABM.** See [Banks and Hooten \(2021\)](#); [Heard et al. \(2015\)](#).
2. Statistical inference
Because of the variety of input rules and the complexity of outputs, the likelihood function of an ABM is generally intractable. One must hence perform likelihood-free inference. [Heard et al. \(2015\)](#) suggest that two main tools allow that: emulators and approximate Bayesian computation (ABC).

Key ideas

Statistics is about reasoning under uncertainty, and therefore **probability distributions**. Inferential statistics learns from data. It asks: *Given sample data, what can we infer about the population?*

In microeconometrics, inference is often conducted under a frequentist approach:

Steps	Options
1. Specify a model intended to approximate the features of the true, unobserved DGP that are relevant to the research question.	(linear regression model w. normal disturbances, logistic regression model, SEM...)
★ Bring in data ★	
2. Estimate the model. In regression analysis, this means estimating the feature of the conditional distribution of $Y \mid X$ specified by the model. Under a parametric specification, this amounts to estimating a parameter vector.	(OLS, GLS, 2SLS, ML,...)
a. Estimation $\implies$ • “$\hat{\beta}^{\text{obs}} = \dots$” • “$\text{CI}_{95\%}[\hat{\beta}]^{\text{obs}} = [\dots, \dots]$”	
b. Hypothesis testing $\implies$ • “H_0 is/isn’t rejected at level α” • “$\hat{\beta}^{\text{obs}}$ is/isn’t statistically significant”	
3. Assess the model’s fit and compare it with alternative models.	

Under frequentist inference, estimators, confidence procedures, and tests are justified by their repeated-sampling properties. These properties may be exact in finite samples or asymptotic (in which case they stem from laws of large numbers and central limit theorems).

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A A small library of regression models

Recall from section 2.3 that a regression model is a statistical model for the **conditional distribution** $Y|X$ or for some feature(s) of that distribution. All regression models are therefore first written as conditional distributions, or features thereof. Only for some of them, depending on the properties of the given distribution, may we also derive a *conditional mean with additive/multiplicative error* form.

In this section, we first describe the textbook classical linear regression model (CLRM), i.e., its assumptions, and its typical estimators with their properties. We then describe other common models that they can be viewed as generalizations of the CLRM along some dimension, such as:

- the power of the independent variables ($\rightarrow$ polynomial regressions);
- the link function relating the linear predictor to the conditional mean ($\rightarrow$ generalized linear models);
- the number of levels in the data ($\rightarrow$ multilevel models)...

We illustrate these generalizations succinctly in the list below, and go into more details in the subsequent subsections. When writing the models as systems of n equations, we consider the indexes $i = 1, \dots, n$ and $j = 1, \dots, n$, $i \neq j$.

• Classical Linear Regression Model

The CLRM is not a fully distributional model but a model of features—namely the two first moments—of $Y|X$. Its errors are conditionally homoskedastic and uncorrelated, but not necessarily i.i.d.³⁴

$$\begin{aligned} \mathbb{E}[Y|X] &= X\beta, & \mathbb{V}[Y|X] &= \sigma^2 I_n \\ \iff Y &= X\beta + u, & \mathbb{E}[u|X] &= 0, & \mathbb{V}[u|X] &= \sigma^2 I_n \end{aligned}$$

• General Linear Regression Model

The general linear regression model similarly models the conditional mean as linear, but the variance-covariance matrix is unrestricted:

$$\mathbb{E}[Y|X] = X\beta$$

• Polynomial Regression

- Ex: LOESS (locally estimated scatterplot smoothing) is a nonparametric regression algorithm, in which $\mathbb{E}[Y_i|X]$ at each data point i is estimated using a weighted low-degree polynomial regression model that gives higher weights to the neighboring points along X .

$$Y_i = \mathbb{E}[Y_i|X] + u_i, \quad \mathbb{E}[Y_i|X] = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \dots + \beta_p X_i^p, \quad u_i|X \sim (0, \sigma_i^2)$$

• Generalized linear model (GLM)

GLMs are often used to predict outcomes of bounded or discrete form (outcomes that cannot be fit well with normally distributed additive errors). A GLM consists of three elements: a probability distribution from the exponential family $\mathcal{D}()_{\text{ExpFamily}}$ we assume the conditional outcome to be generated from,³⁵ a linear predictor $X\beta$, and an invertible link function $g()$ that relates $\mathbb{E}[Y|X]$ to $X\beta$. Examples:

- The linear regression model is a GLM with normal outcome data and identity link.
- The logistic regression model is a GLM with Bernoulli outcome data and logit link.
- The Poisson regression model is a GLM with Poisson outcome data and log link.

$$Y_i|X \sim \mathcal{D}_{\text{ExpFamily}}(\dots), \quad g(\mathbb{E}[Y_i|X]) = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik}$$

³⁴We could also write the stacked model as the distribution $y|X \sim D(X\beta, \sigma^2 I_n)$, but only with the understanding that $D(a, b)$ is an otherwise *unspecified* distribution with mean a and covariance b .

³⁵Distributions in the exponential family have a probability density (or mass) function $f_\theta(y)$ whose form make them highly tractable mathematically. In particular, we can obtain general expressions for their mean and variance in terms of their canonical parameters θ using differentiation. See Wood (2017), section 3.1, on how this allows to develop a general method for fitting a GLM by maximum likelihood, as θ is ultimately determined by the regression parameters.

• Generalized additive model (GAM)

GAMs generalize further to allow for $g(\mathbb{E}[Y|X])$ to depend linearly on smooth (nonlinear) functions of some predictors. Relationships with $X_{i1}, \dots, X_{ik}$ are represented via smooth functions $h_1(\cdot), \dots, h_j(\cdot), \dots$, which allow to estimate flexible non-linear relationships and interaction effects. GAMs can thereby be considered semi-parametric models. They have an additional characteristic: they are fitted using penalized estimation. A penalty for the size of the coefficients for the smooth functions is added to the objective function (whether the likelihood or a loss function), to prevent overfitting the data.

$$Y_i|X \sim \mathcal{D}_{\text{ExpFamily}(\dots)}, \quad g(\mathbb{E}[Y_i|X]) = \beta_0 + h_1(X_{i1}) + \dots + h_k(X_{ik}) + l(X_{i1}, X_{i4}) + \dots$$

• Multilevel aka hierarchical models

The lowest-level model is a regression, higher-level models model coefficients of the model immediately below them, and can be regressions also or mere distributions. All levels are fitted simultaneously. Examples:

- Two-level, varying-intercept model; the group-level is a regression:

$$\begin{cases} Y_i \sim \mathcal{D}(\alpha_{j[i]} + \beta X_i, \sigma_y^2) & \forall i = 1, \dots, n, \quad j = 1, \dots, J \\ \alpha_j \sim \mathcal{D}(\gamma_0 + \gamma_1 W_j, \sigma_\alpha^2) & \forall j = 1, \dots, J \end{cases}$$

- Two-level, varying-intercept & slope model; the group-level models are separate distributions:

$$\begin{cases} Y_i \sim \mathcal{D}(\alpha_{j[i]} + \beta_{j[i]} X_i, \sigma_y^2) & \forall i = 1, \dots, n, \quad j = 1, \dots, J \\ \alpha_j \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) & \forall j = 1, \dots, J \\ \beta_j \sim \Gamma(\gamma_\beta, \delta_\beta) & \forall j = 1, \dots, J \end{cases}$$

• Incomplete data models

- Missing data. For some problems, we can set up a model specifically to handle the missingness mechanism. For example for censored data, extensions of ML or Bayesian regression include the censoring into the likelihood.
- Measurement error in the predictors:³⁶ we observe $X^* = X + \eta$. If we can estimate the variance of the measurement errors, we can either just apply a bias correction on the raw estimate from the regression of Y on X^* , or directly fit the full “simultaneous-equation model” using a marginal likelihood or Bayesian approach. Same maths as in IV.

³⁶Measurement error in y does not pose a problem besides imprecision, as it just goes into the error term. It is measurement error in x that poses a problem: estimated regression coefficients can be attenuated (i.e., it doesn't just increase standard errors, but can drive the coefficient down).

A.1 The classical linear regression model and common estimators

The CLRM is a set of assumptions that describe certain features of the conditional distribution $Y|X$:

Gauss-Markov	(A1)	$Y = X\beta + u$	Linearity and additivity: the systematic component is linear in β .
	(A2)	$X_{n \times p}$ has rank p	Identification No perfect collinearity among the X 's.
	(A3)	$\mathbb{E}[u X] = 0_{n \times 1}$	Strict exogeneity of X $u X$ has no systematic component, as all other factors that affect Y are unrelated to X . ³⁷
	(A4)	$\mathbb{V}[u X] = \sigma^2 I_n$	Conditionally spherical errors
		$\text{cov}[u_i, u_j X] = 0$	– cond. uncorrelated No dependence in the data left unexplained.
		$\mathbb{V}[u_i X] = \sigma^2$	– cond. homoskedastic The conditional variance of the errors is the same across observations and across the support of X .
	(A5)	$u X \sim \mathcal{N}(0, \sigma^2 I_n)$	Normal errors ³⁸

We note that together, assumptions (A1)-(A4) recover the definition of the CLRM given in section 2.3 as the specification of the first two moments of the conditional distribution $Y|X$: $\mathbb{E}[Y|X] = X'\beta$, $\mathbb{V}[Y|X] = \sigma^2$. We also see that estimating the model means estimating its parameters: β and σ^2 .

A.1.1 ML estimator

$\hat{\theta}_{ML} := \arg \max_{\theta} \ln f_{Y|X}(Y|X; \theta)$: By definition, the ML estimator requires a likelihood, i.e., a full distributional assumption for the conditional outcome. To solve for it, we hence assume (A1) to (A5). $Y|X$ is normally distributed, so $f_{Y|X}(Y|X; \beta, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{(Y-X\beta)'(Y-X\beta)}{2\sigma^2}\right)$. The FOCs of the maximization problem then give an exact closed-form solution: $\hat{\beta}_{ML} = (X'X)^{-1}X'Y$; $\hat{\sigma}_{ML}^2 = \frac{r'r}{n}$.

Properties of $\{\hat{\beta}_{ML}, \hat{\sigma}_{ML}^2\}$

$$(A1)-(A5) \implies \begin{cases} \hat{\beta}_{ML} \text{ is } \mathbf{unbiased, efficient,} \text{ and normally distributed.} \\ \hat{\sigma}_{ML}^2 \text{ is downward-biased.}^a \text{ Instead, we can compute the sample variance } s^2 := \frac{r'r}{n-p}. \\ \hat{\beta}_{ML} \text{ and } \hat{\sigma}_{ML}^2 \text{ are } \mathbf{consistent,} \text{ and asymptotically unbiased and efficient.} \end{cases}$$

► The ML estimator has superb statistical properties, but it requires assuming normal errors...

^a $\mathbb{E}[\hat{\sigma}_{ML}^2|X] = \frac{1}{n}\mathbb{E}[r'r|X] = \dots = \frac{n-p}{n}\sigma^2 < \sigma^2$: the variance is under-estimated; but the bias does decrease with n .

A.1.2 OLS estimator

The OLS slope estimator is defined as $\hat{\beta}_{OLS} := \arg \min_{\beta} \sum_{i=1}^n r_i^2$, where r_i denote the residuals from the fit. Under (A1), The FOC of the minimization problem gives an exact closed-form solution: $\hat{\beta}_{OLS} = (X'X)^{-1}X'Y$. The conventional OLS estimator for the population variance of the errors σ^2 is then the bias-adjusted sample variance of the residuals $\hat{\sigma}_{OLS}^2 := s^2 := \frac{r'r}{n-p} = \frac{\sum_i r_i^2}{n-p}$.

³⁷“Strict” means that, for each i , the zero-conditional-mean restriction conditions on all $X_j, \forall j$, not only X_i . Ex: with time series data, this implies that u_t is uncorrelated with X_s for every past, current, and future period s .

³⁸This is not required for estimating the regression but for making sampling-based inference, *when n is small*. Indeed, it is used to justify the usual t - and F -based inference exactly in finite samples. If $u|X \sim \mathcal{N}(0, \sigma^2 I)$, then $\hat{\beta}_{OLS}|X$ is normally distributed and the Studentized statistic $\frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)}$ has an exact Student $\mathcal{T}_{n-p}$ distribution, which we can hence use to perform a t -test. When n is large enough, laws of large numbers and central limit theorems say that the *asymptotic* sampling distribution of $\hat{\beta}_{OLS}$ is normal, regardless of the errors. However, with small n , the sampling distribution of $\hat{\beta}$ does depend on the distribution of the errors, and if they aren't normal, then $\hat{\beta}_{OLS}$ isn't normal. One may then want to consider an alternative, e.g., bootstrap.

Properties of $\{\hat{\beta}_{\text{OLS}}, \hat{\sigma}_{\text{OLS}}^2\}$

$$\begin{aligned}
\text{(A1)-(A3)} &\implies \begin{cases} \hat{\beta}_{\text{OLS}} \text{ is } \textbf{unbiased}. \\ \hat{\beta}_{\text{OLS}} \text{ is } \textbf{consistent}, \text{ and asymptotically unbiased, efficient, and normally distributed.} \\ \hat{\sigma}_{\text{OLS}}^2 \text{ is } \textbf{consistent}, \text{ and asymptotically unbiased and efficient.} \end{cases} \\
+ \text{(A4)} &\implies \hat{\beta}_{\text{OLS}} \text{ is } \textbf{efficient among linear unbiased estimators.}^a \\
+ \text{(A5)} &\implies \hat{\beta}_{\text{OLS}} = \hat{\beta}_{\text{ML}} \text{ is } \textbf{efficient}, \text{ and normally distributed.}
\end{aligned}$$

► We see why OLS is the canonical estimator for linear regression: it is unbiased as long as regressors are properly exogenous—i.e., notably regardless of potential structure remaining in the errors!

^aThis is the Gauss-Markov Theorem: “If the Gauss-Markov assumptions are satisfied, then the OLS estimator is the best linear unbiased estimator (BLUE)”. Here, linearity refers to the linearity of $\hat{\beta}$ w.r.t. the vector Y .

Inference requires a $\hat{\mathbb{V}}[\hat{\beta}]$ After estimation, we typically pursue sampling-based statistical inference, such as statistical tests and the computation of confidence intervals. They involve a standard error, i.e., an estimator of $\hat{\beta}$ ’s standard deviation. We hence need an estimator of $\hat{\beta}$ ’s sampling variance-covariance matrix. There are multiple valid choices, below we describe first those that rely on asymptotics—namely, the conventional OLS and sandwich estimators—and an alternative: bootstrap.

• Asymptotics

Assuming (A1)-(A3), by applying a CLT, we obtain the limit distribution of the rescaled $\hat{\beta}_{\text{OLS}}$:³⁹

$$\begin{aligned}
\sqrt{n}(\hat{\beta}_{\text{OLS}} - \beta) &= \left(\frac{1}{n}X'X\right)^{-1} \frac{1}{\sqrt{n}}X'u = \underbrace{\left(\frac{1}{n} \sum_i X_i X_i'\right)^{-1}}_{\xrightarrow[n \rightarrow \infty]{p} M_{\text{XX}}} \underbrace{\frac{1}{\sqrt{n}} \sum_i X_i u_i}_{\xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, M_{\text{X}\Omega\text{X}})} \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}\left(0, M_{\text{XX}}^{-1} M_{\text{X}\Omega\text{X}} M_{\text{XX}}^{-1}\right)
\end{aligned}$$

- where
- $M_{\text{XX}} := \text{plim}\left(\frac{1}{n}X'X \mid X\right) = \lim\left(\mathbb{E}\left[\frac{1}{n}X'X \mid X\right]\right) = \lim\left(\frac{1}{n}X'X\right)$ is finite, symmetric, and $\neq 0_n$
 - $M_{\text{X}\Omega\text{X}} := \text{plim}\left(\frac{1}{n}X'uu'X \mid X\right) = \lim\left(\mathbb{E}\left[\frac{1}{n}X'uu'X \mid X\right]\right) = \lim\left(\frac{1}{n}X' \mathbb{E}[uu'|X] X\right) =: \lim\left(\frac{1}{n}X'\Omega X\right)$
 - Ω is the variance-covariance matrix of the error term: $\mathbb{E}[uu'|X]$

We write the limit distribution of $\sqrt{n}(\hat{\beta}_{\text{OLS}} - \beta)$ and not $\hat{\beta}_{\text{OLS}}$, as $\hat{\beta}_{\text{OLS}}$ has a degenerate distribution with all mass at β . However, for convenience, we would like to think of the distribution of $\hat{\beta}_{\text{OLS}}$. We do so by introducing the artifice of “asymptotic distribution”. We consider n large but not infinite, such that the asymptotics have kicked in, and we can drop the limits in mathematical expressions (lim is dropped, plim becomes $\mathbb{E}$). We obtain $\hat{\beta}_{\text{OLS}}$ ’s asymptotic distribution:

$$\hat{\beta}_{\text{OLS}} \overset{\text{a}}{\sim} \mathcal{N}\left(\beta, \frac{1}{n} \left(\frac{1}{n}X'X\right)^{-1} \left(\frac{1}{n}X'\Omega X\right) \left(\frac{1}{n}X'X\right)^{-1'}\right) = \mathcal{N}\left(\beta, \underbrace{(X'X)^{-1} X'\Omega X (X'X)^{-1'}}_{\overset{\text{a}}{\mathbb{V}}[\hat{\beta}_{\text{OLS}}]}\right)$$

^a $\mathbb{V}[\hat{\beta}_{\text{OLS}}]$ is unknown as Ω is unknown. We need a **consistent** estimator of it, either directly as a whole, or of Ω , to plug in. The relevant estimator for consistency will hence depend on the error structure.

► Spherical (A4)

If assumption (A4) is satisfied, then $\overset{\text{a}}{\mathbb{V}}$ simplifies to $\sigma^2 (X'X)^{-1}$. The population variance σ^2 is consistently estimated by the conventional OLS bias-adjusted sample variance s^2 , and hence $\overset{\text{a}}{\mathbb{V}}$ by $\overset{\text{a}}{\hat{\mathbb{V}}}_s := s^2 (X'X)^{-1}$. As this is the Cramer-Rao lower bound, $\hat{\beta}_{\text{OLS}}$ is efficient.

³⁹For convenience, we drop the notation $|X$, however all features of $\hat{\beta}_{\text{OLS}}$ ’s distribution are actually conditional on X .

► **Not spherical**

If assumption (A4) is violated—due to heteroscedasticity or dependence—we need a covariance matrix estimator that is consistent under this misspecification of the remaining likelihood. One approach is to use **sandwich estimators**.⁴⁰ We decompose the variance into its $3 \times k$ components: *bread*, *meat*, *bread*, and compute a consistent estimator of the *meat* component $X'\Omega X$ that best represents our assumed error structure, to finally compute:

$$\hat{\mathbb{V}}^{\hat{a}}[\hat{\beta}_{\text{OLS}}] := \underbrace{(X'X)^{-1}}_{\text{bread}} \underbrace{X'\hat{\Omega}X}_{\text{meat}} \underbrace{(X'X)^{-1'}}_{\text{bread}}$$

Why not use a GLS estimator directly? The first-best strategy would be to calculate an efficient estimator directly, instead of accepting an inefficient OLS and adjusting the SEs ex-post. Here, that estimator is GLS. However, its efficiency rests on the correct specification of the heteroskedasticity and/or autocorrelation pattern. I.e., GLS is the best approach if we know Ω , but we usually don't.

• **Heteroskedastic**

$\Omega = \text{diag}[\sigma_i^2]$. White (1980) proposes to use $\hat{\Omega}_H := \frac{1}{n-p} \text{diag}[r_i^2]$, i.e., $X'\hat{\Omega}_H X = \frac{1}{n-p} \sum_i r_i^2 X_i X_i'$.

The resulting non-parametric estimator $\hat{\mathbb{V}}_H^{\hat{a}}$ is consistent for $\mathbb{V}^{\hat{a}}$, even though r_i^2 is inconsistent for σ_i^2 . The resulting standard errors are called heteroskedasticity-consistent, aka “robust”. Robust SEs seem to have become best practice with large samples, as one can rarely assume homoskedastic errors.^{41,42}

• **Autocorrelated**

If errors are autocorrelated in any way (in time, in space, by groups...), it means that the model is not capturing some feature of the DGP. To conduct proper inference, one can either treat this structure as *substance* and incorporate it in the model (e.g., if errors are autocorrelated by group, by modeling a multilevel data structure), or treat it as *nuisance* and adjust for it after fitting the model (e.g., if errors are autocorrelated by group, by clustering standard errors).

In the second approach, the dependence structure is left in the errors, such that the true covariance matrix of the errors Ω has some non-diagonal terms that are non-zero. We need an estimator $\hat{\Omega}$ that is consistent for all diagonal and non-diagonal terms. The sandwich estimators below achieve this with a common approach:

- They assume the process is 2nd order stationary;⁴³
- They use sample equivalents: White’s estimates for diagonal terms (the estimators are hence also heteroskedasticity-consistent), and weighted (e.g., through a kernel) sample autocovariances for non-diagonal covariance terms.

⁴⁰All extremum estimators can actually be shown to be consistent and asymptotic normal, with an asymptotic variance matrix in the sandwich form: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, A(\theta)^{-1}B(\theta)A(\theta)^{-1})$. The sandwich algorithm presented here for OLS can be extended to all extremum estimators, e.g., ML and GMM. This is not to say that it should be; Freedman (2006) points out that while White’s sandwich estimator often gives good results in OLS, the equivalent correction in ML does not necessarily make sense: “If the model is nearly correct, so are the usual standard errors, and robustification is unlikely to help much. On the other hand, if the model is seriously in error, the sandwich may help on the variance side, but the parameters being estimated by the ML are likely to be meaningless.” If the specification—and hence the likelihood function—is incorrect, the parameter will be biased; why care about the variance of an estimator for the wrong parameter?

⁴¹For example, using a linear model to approximate a nonlinear conditional expectation function (CEF) should lead to heteroskedasticity (Angrist and Pischke, 2008, p.35). Indeed, as the quality of fit between the regression line and the CEF varies with X_i , the residuals are larger, on average, at values of X_i where the fit is poorer. The residual variance increases with the square of the gap between the regression line and the CEF.

⁴²Robust SEs are not necessarily larger than conventional standard errors under heteroskedasticity. Robust SEs estimate the variance using the observation-specific error variances rather than incorrectly replacing them all by one common $\hat{\sigma}^2$. What matters is where the high-variance observations lie in regressor space: if the most influential observations for estimating a given coefficient also have relatively large error variances, the conventional SE may be too small; if on the contrary they have relatively small error variances, the conventional SE may be too large.

⁴³A stationary process is a stochastic process whose unconditional joint probability distribution does not change over the dimension of the process. I.e., here:

- for a time series: the correlation between 2 observations that are m periods apart is the same across the period;
- for a spatial process: the correlation between 2 observations that are a distance d apart is the same across the spatial field;
- for a process across groups: the correlation between 2 observations is fully determined by their group appartenance.

They are thereby fully non-parametric, and account for dependence of unknown form along the dimension of autocorrelation.

◦ **Clustering (dimension of autocorrelation: group appartenance)**

Errors are correlated within G groups or “clusters”: $\mathbb{E}[u_i|X] = 0$, $\mathbb{E}[e_i e_j|X] \neq 0$, $\forall i, j \in \text{same group } g$. The covariance matrix of the error term Ω has a block-diagonal structure.

The *cluster-robust* estimator uses $X' \widehat{\Omega}_C X = \frac{1}{n-p} \sum_{g=1}^G X'_g r_g r'_g X_g$.

Δ This estimator is valid only if G is sufficiently large (rule of thumb: > 30), as, like all sandwich estimators, it relies on asymptotics. Intuitively, fewer clusters means there is less independent information in the sample. And if clusters are unbalanced, the effective number of clusters is even lower. [Cameron and Miller \(2015\)](#) recommends using critical values from the Student's $\mathcal{T}_{G-1}$ distribution instead of from the standard normal $\mathcal{N}(0, 1)$.

◦ **Serial correlation (dimension of autocorrelation: time)**

[Newey and West \(1987\)](#) proposes an estimator that accounts for serial correlation of unknown form in an error time series $\{u_t\}_t$. It can be expanded to panel datasets by estimating only correlations between lagged errors within the same cluster.

The time series' autocovariance of lag l is $\gamma_u(t, t-l) := \text{cov}[u_t, u_{t-l}] = \mathbb{E}[(u_t - \mu_{u_t})(u_{t-l} - \mu_{u_t})]$. If the process is covariance-stationary, it is a function of the relative lag only: $\gamma_u(l)$. Its bias-adjusted sample equivalent, i.e., the sample autocovariance, is $g(l) = \frac{1}{T-k} \sum_{t=l+1}^T r_t r_{t-l}$.

The Newey-West estimator weights these sample autocovariances using a triangular kernel function, s.t. the weight decreases linearly with the lag up to a chosen maximum lag L ,⁴⁴ and adds White's variance estimates:

$$\begin{aligned} \widehat{\Omega}_{\text{NW}} &:= G(0) + \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) [G(l) + G(l)'] \\ \implies X' \widehat{\Omega}_{\text{NW}} X &:= \frac{1}{T-k} \sum_{t=1}^T r_t^2 X_t X_t' + \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) \left[\frac{1}{T-k} \sum_{t=l+1}^n r_t r_{t-l} X_t X_{t-l}' + \frac{1}{T-k} \sum_{t=l+1}^n r_t r_{t-l} X_{t-l} X_t' \right] \\ &= \frac{1}{T-k} \sum_{t=1}^T r_t^2 X_t X_t' + \frac{1}{T-k} \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) \sum_{t=l+1}^n r_t r_{t-l} (X_t X_{t-l}' + X_{t-l} X_t') \end{aligned}$$

◦ **Spatial correlation (dimension of autocorrelation: space)**

This dimension is actually a dual dimension: while time or group appartenance are 1D, space is at least 2D. [Conley \(1999\)](#), under the supplementary assumption that the process is isotropic, proposes a consistent estimator for Ω that accounts for spatial correlation of unknown form in the errors. It weights the sample covariances using a kernel function $k(s_i, s_j)$, where s_i is the location of observation i .⁴⁵

$$X' \widehat{\Omega}_{\text{Co}} X := \frac{1}{n-p} \sum_{i=1}^n r_i^2 X_i X_i' + \frac{1}{n-p} \sum_{i=1}^n \sum_{j=1}^n k(s_i, s_j) r_i r_j X_i X_j'$$

• **Block bootstrap**

⁴⁴ $\widehat{\Omega}$ is consistent iff $L \rightarrow \infty$ and $\frac{L}{T^{1/4}} \rightarrow 0$ as $T \rightarrow \infty$, i.e., iff L grows slower than $T^{1/4}$. A common practice is hence to set L to the integer part of $T^{1/4}$.

⁴⁵Multiple choices of kernel are possible. [Conley \(2008\)](#) presents the uniform kernel but does not recommend it over another. $\widehat{\Omega}$ will be consistent if $\forall h, k(s, s+h) \rightarrow 1$ as $n \rightarrow \infty$, but slowly enough for the variance of $\widehat{\Omega}$ to collapse to zero. We need a reasonable effective number of independent clusters, which decreases as the radius is extended. Assuming a stationary and isotropic process, $k(i, j)$ simplifies to a function of distance: $k(d_{ij})$. One can choose whichever distance metric fits one's context, e.g., a metric of economic distance. Note also that spatial dependence does not imply that Conley SEs will necessarily be higher than spherical SEs. In [Conley \(1999\)](#)'s empirical example, 6 out of 9 spatial SE estimates are smaller than their iid counterparts.

Using an estimator of the asymptotic covariance matrix may be difficult (e.g., when the dependence structure is difficult to derive analytically) or unreasonable (when the sample size or the number of clusters is too small to appeal to asymptotics).

Instead, we can use bootstrap methods to approximate the sampling distribution of $\hat{\beta}$ empirically, by repeated resampling with replacement that reproduces the dependence structure.

Consider the case of n observations with cluster-dependence.

- The main assumption of bootstrap is that the data represent the actual underlying (population) distribution well, and hence that a random sample drawn from a random sample from the population is also a random sample of the population itself.
- ☞ An ordinary bootstrap (resampling individual observations) implicitly treats observations as iid units, and hence would not be able to replicate the dependence structure in the data.
- ☞ Instead, in block bootstrap, we draw samples by resampling entire blocks of correlated observations, which preserves the local dependence approximately.

Method:

1. Draw B bootstrap samples by randomly selecting J clusters with replacement (so some clusters will be selected more than once and others not at all), and including all observations within these clusters. As a result, the bootstrapped sample (b) may not be of size n , which is fine.
2. Refit the model on each sample.
3. Obtain a distribution $\hat{\beta}^{*(1)}, \dots, \hat{\beta}^{*(B)}$
4. Compute the variance of this approximating distribution:

$$\hat{\mathbb{V}}^{\text{boot}}[\hat{\beta}] := \frac{1}{B-1} \sum_{b=1}^B (\hat{\beta}^{*(b)} - \bar{\beta}^*) (\hat{\beta}^{*(b)} - \bar{\beta}^*)'$$

A.2 Generalized linear models

Generalized linear models (GLMs) are often used to predict outcomes of limited form, i.e., that are categorical or constrained to fall in a certain range. With such data, linear regression estimation is not appropriate as it does not take into account the constraint on values of the dependent variable. A common strategy is to transform the limited Y using a link function $g(\cdot)$ into a continuous, real-valued variable $g(Y) \in (-\infty, \infty)$, that we can then model as $g(Y) = X'\beta + u$.

A GLM consists of three elements: a probability distribution $\mathcal{D}_{\text{ExpFamily}(\dots)}$ from the exponential family for the conditional outcome, a linear predictor $X'\beta$, and an invertible link function $g(\cdot)$ that relates it to $\mathbb{E}[Y|X]$:

$$Y|X \sim \mathcal{D}_{\text{ExpFamily}(\dots)}, \quad g(\mathbb{E}[Y|X]) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$$

GLMs encompass for example the following appropriate regression models given the type of Y :

- binary $Y \in \{0, 1\}$: probit regression, logit regression
- count $Y \in \{0, 1, 2, 3, \dots\}$: Poisson regression, negative binomial regression
- interval $Y \in [0, 1]$: fractional response
- censored Y : Tobit regression

A.2.1 Binary outcome models

The outcome variable is binary, i.e., it follows a Bernoulli distribution:

$$Y_i|X_i \sim \text{Ber}(\pi_i) := \begin{cases} 1 & \text{with probability } \pi_i \\ 0 & \text{with probability } 1 - \pi_i \end{cases} \quad \forall i = 1, \dots, n$$

The conditional mean $\mathbb{E}[Y_i|X_i]$ is equal to the conditional probability $\pi_i := P(Y_i=1|X_i)$.⁴⁶ A regression model is therefore formed by expressing π as a function of X_i and β ;⁴⁷ and we look for a link function $g(\cdot)$ that maps the $[0,1]$ interval to the real line.

$$Y_i|X_i \sim \text{Ber}(\pi_i), \quad \pi_i = \mathbb{E}[Y_i|X_i] = g^{-1}(X_i; \beta)$$

• Linear probability model

$$Y_i|X_i \sim \text{Ber}(\pi_i), \quad \pi_i = \mathbb{E}[Y_i|X_i] = X_i'\beta$$

This model is probably the first one that comes to mind. It is not appropriate, as the identity link is not a CDF, it will not constrain the predicted values to be in $[0,1]$, since the predictor $X_i'\beta$ can take any real value. Yet, it is still frequently preferred to Logit or Probit, on grounds that it is computationally simpler, the estimated marginal effects are easier to interpret, and are usually very similar anyway, especially with a large sample size.

However, [Horrace and Oaxaca \(2006\)](#) show that in almost all circumstances, the LPM yields biased and, most importantly, *inconsistent* estimates. I.e., the LPM gives the wrong answer, with almost certainty, even with an infinitely large sample: “consistency seems to be an exceedingly rare occurrence as one would have to accept extraordinary restrictions on the joint distribution of the regressors. Therefore, OLS is frequently a biased estimator and almost always an inconsistent estimator of the LPM.”

• Logit model = Log-odds model = Logistic regression model

$$Y_i|X_i \sim \text{Ber}(\pi_i), \quad \pi_i = \text{logit}^{-1}(X_i'\beta) = \frac{e^{X_i'\beta}}{1 + e^{X_i'\beta}} \iff \text{logit}(\pi_i) = X_i'\beta$$

We choose as link function $g(\cdot)$ the logit $\text{logit}(\cdot) := \ln\left(\frac{\cdot}{1-\cdot}\right)$ (i.e., we choose as $g^{-1}(\cdot)$ the CDF of the logistic distribution: $\text{logit}^{-1}(\cdot)$), which maps $[0,1]$ to $[-\infty, \infty]$. We transform the probability outcome using this logit or “log-odds” transformation. As this new outcome need not be in $[0,1]$, we can model it as a *linear* function of the covariates.

⁴⁶ As $\mathbb{E}[Y|X] = 1 \times P(Y=1|X) + 0 \times P(Y=0|X) = P(Y=1|X)$.

⁴⁷ The function $g^{-1}(\cdot)$ should be a *cumulative distribution function*, to ensure that $0 \leq \pi_i \leq 1$.

The raw slope estimates are in log-odds units, which are often difficult to interpret. One can easily convert them to either the odds scale or the probability scale. Interpretation of each coefficient estimate $\hat{\beta}_j$ is hence, for the example of a continuous X_j , as “a 1-unit difference in X_j is associated with... (keeping the other predictors constant)”:

- logit scale $[-\infty, \infty]$...an average $\hat{\beta}_j$ -unit difference in log-odds($Y=1$)
- odds⁴⁸ scale $[0, \infty]$...an average $e^{\hat{\beta}_j}$ multiplicative difference in odds($Y=1$)
- probability scale $[0, 1]$...a $\frac{\hat{\beta}_j}{4}$ -unit maximum⁴⁹ difference in $P(Y=1)$

• Probit model

$$Y_i|X_i \sim \text{Ber}(\pi_i), \quad \pi_i = \text{probit}^{-1}(X_i'\beta) := \int_{-\infty}^{X_i'\beta} \phi(t)dt \iff \text{probit}(\pi_i) = X_i'\beta$$

We choose as link function $g(\cdot)$ the probit, which is the quantile function of the standard normal distribution (i.e., we choose as $g^{-1}(\cdot)$ the CDF of the normal distribution: $\text{probit}^{-1}(\cdot)$), which maps $[0, 1]$ to $[-\infty, \infty]$.

We cannot interpret each coefficient $\hat{\beta}_j$ directly, we need to compute the marginal effects.

Note: As a rule of thumb, probit regression coefficients are roughly equal to logistic regression coefficients divided by 1.6.

Estimation by Maximum Likelihood $Y|X \sim \text{Ber}(\pi_i)$, hence the conditional density of each observation i is: $f_{Y|X}(y_i|x_i) = \pi_i^{y_i}(1 - \pi_i)^{(1-y_i)}$. Assuming independence across i , the log-likelihood of the data is then the log-likelihood for n independent Bernoulli observations:

$$\begin{aligned} \hat{\theta}_{\text{ML}} &= \arg \max_{\theta} \log \mathcal{L}(y|X; \theta) = \arg \max_{\theta} \log \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \\ &= \arg \max_{\theta} \sum_{i=1}^n y_i \ln(\pi_i) + (1 - y_i) \ln(1 - \pi_i) \\ &= \arg \max_{\beta} \sum_{i=1}^n y_i \ln(g^{-1}(x_i'\beta)) + (1 - y_i) \ln(1 - g^{-1}(x_i'\beta)) \end{aligned}$$

Δ Don't fit logistic models for binary outcomes when the underlying continuous variable is available. For inference or prediction, it is much more efficient to model the underlying continuous variable and then map it back to the probability of the discrete outcome. Ex:

- basketball game: model the expected score differential, and then map it to $\text{Pr}(\text{winning})$.
- elections: predict vote differential and then map that to $\text{Pr}(\text{winning})$.
- health: model change in blood pressure, and then convert it to the binary disease state $\text{Pr}(\text{hypertension})$.

A.2.2 Count data models

Suppose a count outcome $Y_i \in \{0, 1, 2, \dots\}$, i.e., the number of occurrences of an event. Ex: number of children in a household, number of doctor visits per year, number of new cases of an infectious disease per day...

• Poisson regression model

The Poisson distribution $\text{Pois}(\lambda)$ models the number of events occurring in a fixed interval (of time or space), when these events occur at random, independently in time, with the constant mean rate λ . Its probability mass function is therefore $\text{Pr}(Y=y) = \frac{e^{-\lambda} \lambda^y}{y!}$, which further implies $\mathbb{E}[Y] = \mathbb{V}[Y] = \lambda$. The general Poisson regression model is:

$$Y_i|X_i \sim \text{Pois}(\lambda_i), \quad \lambda_i = g^{-1}(X_i'\beta)$$

⁴⁸The odds of success are defined as the ratio of the probability of success π over the probability of failure. Here, where “success” is $Y=1$, the odds of $Y=1$ are $\frac{\pi}{1-\pi}$ to 1.

⁴⁹ Δ The logistic function $\text{logit}^{-1}(\cdot)$ is nonlinear, so the expected difference in $\text{Pr}(Y=1)$ from a given difference in X is not a constant along X . We must choose where to evaluate changes, if we want to interpret them on the probability scale. The slope of the logistic regression curve is steepest at its halfway point ($\text{logit}^{-1}(\cdot) = 0.5$), where it equals $\beta/4$. I.e., the largest change in π from a 1-unit change in X is $\beta/4$.

A common choice of link function $g()$ is $\ln()$. The Poisson regression model is therefore fitted as a log-linear regression with Poisson error distribution: $Y_i|X_i \sim \text{Pois}(e^{X_i'\beta})$.

Estimation by Maximum Likelihood:

$$\begin{aligned}\hat{\beta}_{\text{ML}} &= \arg \max_{\beta} \log \mathcal{L}(Y | X; \beta) = \arg \max_{\beta} \log \prod_{i=1}^n \Pr(Y_i=y_i | X_i=x_i; \beta) \\ &= \arg \max_{\beta} \sum_{i=1}^n \log \frac{e^{-e^{X_i'\beta}} (e^{X_i'\beta})^{y_i}}{y_i!} \\ &= \arg \max_{\beta} \sum_{i=1}^n \left[-e^{X_i'\beta} + Y_i(X_i'\beta) - \ln(y_i!) \right]\end{aligned}$$

A limitation of the Poisson model is that it implies equi-dispersion, i.e., that the variance is equal to the mean: $\mathbb{V}[Y_i|X_i] = \mathbb{E}[Y_i|X_i]$, whereas we almost always see overdispersion in the data (ex: a few traders will do many trades, many traders will do a few). To accomodate overdispersion, some softwares (e.g., R) have packages that permit an “adjusted” Poisson regression, or, preferably, we can turn to the negative binomial distribution.

• Negative binomial model

The negative binomial distribution $\text{NB}(p, r)$ models the number of successes in a sequence of iid Bernoulli(p) trials before r failures occur. Its probability mass function is $\Pr(Y=y) = \binom{y+r-1}{y} p^y (1-p)^r = \frac{\Gamma(y+r)}{y! \Gamma(r)} p^y (1-p)^r$.

The NB allows the variance to be larger than the mean, which makes it a useful overdispersed alternative to the Poisson.

In a regression framework, it is more intuitive to specify the distribution in terms of its mean $\mu = \frac{pr}{1-p}$ and the reciprocal overdispersion parameter r . We can rewrite the probability mass function as $\Pr(Y=y) = \frac{\Gamma(y+r)}{y! \Gamma(r)} \left(\frac{\mu}{r+\mu}\right)^y \left(\frac{r}{r+\mu}\right)^r$. The distribution converges to Poisson as $r \rightarrow \infty$, i.e., as the overdispersion $\frac{1}{r} \rightarrow 0$.

Using as link function $g()$ the usual logarithmic transformation $\ln()$, the NB regression model is:

$$Y_i|X_i \sim \text{NB}(\mu_i, r), \quad \mu_i = e^{X_i'\beta}$$

Estimation by Maximum Likelihood:

$$\begin{aligned}\hat{\theta}_{\text{ML}} &:= \arg \max_{\theta} \log \mathcal{L}(Y|X, \theta) = \arg \max_{\theta} \log \prod_{i=1}^n \Pr(Y_i=y_i | X_i=x_i; \theta) \\ &= \arg \max_{\theta} \sum_{i=1}^n \log \left(\frac{\Gamma(y_i + r)}{y_i! \Gamma(r)} \left(\frac{\mu_i}{r + \mu_i} \right)^{y_i} \left(\frac{r}{r + \mu_i} \right)^r \right) \\ &= \arg \max_{\beta, r} \sum_{i=1}^n \log \left(\frac{\Gamma(y_i + r)}{y_i! \Gamma(r)} \left(\frac{e^{X_i'\beta}}{r + e^{X_i'\beta}} \right)^{y_i} \left(\frac{r}{r + e^{X_i'\beta}} \right)^r \right)\end{aligned}$$

Interpretation of each coefficient $\hat{\beta}_j$ (keeping all the other predictors fixed):⁵⁰

- log scale $[-\infty, \infty]$ “a 1-unit difference in X_j corresponds to a $\hat{\beta}_j$ -unit difference in $\log(\mu)$.”
- incidence rate ratio scale $[0, \infty]$ “a 1-unit difference in X_j corresponds to a $e^{\hat{\beta}_j}$ multiplicative difference in μ .”

Example: Y_i := the number of new cases of an infectious disease on day t_i . In the early stages of an epidemic, the rate of new cases can increase exponentially, s.t. $\mathbb{E}[Y_i|X_i] = \gamma \exp(\delta \cdot t_i)$ is a suitable model. Using a log link turns the model into a GLM: $\log(\mathbb{E}[Y_i|X_i]) = \log(\gamma) + \delta \cdot t_i$. The complete specification of a reasonable GLM consists therefore in the log link, the linear predictor $\beta_0 + \beta_1 \cdot t_i$, and a Poisson or negative binomial outcome distribution.

⁵⁰In R, The `jtools` package provides an easy way to exponentiate the coefficients and get the corresponding CIs.

A.3 Generalized additive models

We want to capture relationships between the response and predictors that are potentially non-linear. Generalized additive models (GAMs) let $g(\mathbb{E}[Y_i|X_i])$ depend on smooth (nonlinear) functions $h_1(), h_2(), \dots$ of the predictors, where each such smooth term is made of a sum of “basis functions”, as is described further below. The overall GAM is thereby a linear model in transformed variables, such that fitting can proceed with similar techniques as for linear models. These smooths enable to capture non-linear relationships and interaction effects with a high degree of flexibility; GAMs are a form of semi-parametric models.

Specification We consider a simple GAM with parametric terms $X_p\theta$, a univariate smooth term $h(X_1)$, and a bivariate smooth term $l(x_2, x_3)$, and describe the forms that the smooths can take:

$$Y_i|X_i \sim \mathcal{D}_{\text{ExpFamily}}(\dots), \quad g(\mathbb{E}[Y_i|X_i]) = X_{ip}\theta + h(X_{i1}) + l(X_{i2}, X_{i3})$$

- **Univariate smooth $h()$**

$h(X)$ is made of a linear basis expansion in x , i.e., of a sum of “basis functions” $b_j()$ s, thus named for they belong to a common space of functions or “basis”: $h(X) = \sum_{j=1}^k b_j(X)\beta_j$. Examples of bases include:

- **Polynomial basis**

$$b_1(X) = 1, \quad b_2(X) = X, \quad b_3(X) = X^2, \quad \dots \implies h(X) = \beta_1 + \beta_2 X + \beta_3 X^2 + \dots$$

- **Piecewise bases**

The range of X is divided into multiple intervals at “knots” $x_1^*, \dots, x_k^*$, and piecewise functions are fit on the intervals. The number of knots k , sometimes called the dimension of the piecewise function, controls the degree of flexibility. Examples of basis functions $\{b_j\}_1^k$:

- * **Piecewise constant basis**

$b_1(X) = \mathbb{1}\{X < x_1^*\}$, $b_2(X) = \mathbb{1}\{x_1^* \leq X < x_2^*\}$, ..., $b_k(X) = \mathbb{1}\{x_{k-1}^* \leq X\}$. The resulting $h()$ is a step function.

- * **Piecewise linear basis**

Piecewise linear functions are fit on the intervals.

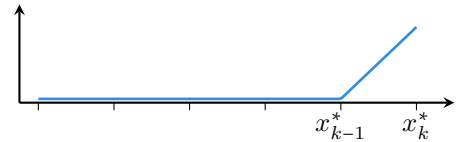
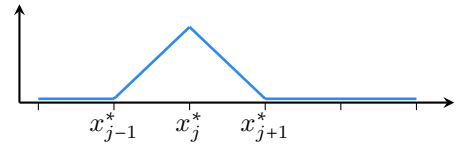
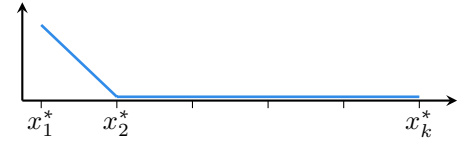
$$b_1(X) = \begin{cases} (x_2^* - X)/(x_2^* - x_1^*) & \text{if } X < x_2^* \\ 0 & \text{otherwise} \end{cases}$$

...

$$b_j(X) = \begin{cases} (X - x_{j-1}^*)/(x_j^* - x_{j-1}^*) & \text{if } x_{j-1}^* < X < x_j^* \\ (x_{j+1}^* - X)/(x_{j+1}^* - x_j^*) & \text{if } x_j^* < X < x_{j+1}^* \\ 0 & \text{otherwise} \end{cases}$$

...

$$b_k(X) = \begin{cases} 0 & \text{if } X \leq x_{k-1}^* \\ (X - x_{k-1}^*)/(x_k^* - x_{k-1}^*) & \text{otherwise} \end{cases}$$



- * **Piecewise polynomial basis with continuous derivatives = spline**

A spline is a piecewise polynomial function of degree m , with the imposed constraint of $m - 1$ continuous derivatives, to ensure that the pieces join smoothly. Examples:

- A linear spline is a piecewise linear polynomial continuous at each knot. It has $2(k+1) - k = k + 2$ degrees of freedom.
- A cubic spline is a piecewise cubic polynomial with continuous 1st and 2nd derivatives at each knot.⁵¹ The set of cubic splines with fixed knots is a vector space with $4(k+1) - 3k =$

⁵¹The cubic spline $h()$ can be shown to be the smoothest interpolator for any given set of points $\{x_i, y_i : i = 1, \dots, n\}$, i.e., $h() = \arg \min_{f()} \int_{x_1}^{x_n} f''(x)^2 dx$, where $f()$ are functions continuous on $[x_1, x_n]$ with continuous 1st derivatives. As a cubic spline can closely approximate any underlying smooth function, there is seldom any good reason to use a higher-order polynomial.

$k + 4$ degrees of freedom. Ex: a cubic spline with a single knot at a point c :

$$Y_i = \begin{cases} \beta_{01} + \beta_{11}X_i + \beta_{21}X_i^2 + \beta_{31}X_i^3 + \epsilon_i & \text{if } X_i < c \\ \beta_{02} + \beta_{12}X_i + \beta_{22}X_i^2 + \beta_{32}X_i^3 + \epsilon_i & \text{if } X_i \geq c \end{cases}$$

Note: Polynomials tends to be erratic near the boundaries. To prevent that behavior, one can use a “natural” or “restricted” spline, which further imposes linearity beyond the boundary knots. E.g., a natural cubic spline sets the cubic and quadratic terms there to 0. This frees up 4 degrees of freedom (2 at each end of the curve), moving the total from $k + 4$ to k .

Knot selection or penalization Piecewise bases require choosing the number and location of knots. There is a trade-off between a high enough number of knots to capture non-linearities, but not so many that we’re overfitting. How to select them? One could place knots either at uniform quantiles of the data, or in regions of the support where more variation in the relationship is expected, and compare different numbers of knots by cross-validation... An alternative method to avoid the knot selection problem entirely is penalization: we set a very high number of knots k , but add to the objective function (whether the likelihood or a loss function) a penalty for the ‘wiggleness’ or curvature of the smooth, to prevent overfitting the data. The objective function becomes:

* in maximum likelihood estimation, a penalized likelihood: $l_p(\beta) := l(\beta) - \text{penalty}$

* in least squares estimation, a penalized least squares: $\text{Loss} = \sum_{i=1}^n (Y_i - h(X_i))^2 + \text{penalty}$

with penalty $:= \lambda \int [h''(x)]^2 dx = \lambda \beta' S \beta$. The smoothing parameter λ establishes a tradeoff between the first usual term of the objective function, which measures closeness to the data, and the second term which penalizes curvature in the function (captured by its second derivative). Increasing λ makes the function smoother, with $\lambda \rightarrow \infty$ corresponding to a straight line fit. The penalty can also be expressed in a quadratic form using a penalty matrix S .

While it may seem that the model is over-parameterized, due to the high number of knots, i.e., of basis functions, the penalty term actually translates to a penalty on the magnitude of the spline coefficients, which are shrunk toward the linear fit, reducing the effective degrees of freedom.⁵² The choice of the basis dimension is therefore not critical, as the smoothing parameters λ_j control the actual model complexity, as long as k is not set to be too small (which will force oversmoothing).⁵³

• Multivariate smooth $l(W, Z)$

We can also construct smooths of multiple variables.

- For isotropic smooths, i.e., functions that aim to achieve the same smoothness per unit change in W and Z , one can use a thin plate spline — not detailed here.
- When it is unreasonable to impose the same smoothness per unit change in W and Z (e.g., when the covariates are measured in fundamentally different units, such that it is difficult to scale them relative to one another), one can use a **tensor product smooth**. A tensor product is constructed from ‘marginal smooths’ of single covariates, each with its own basis and associated quadratic penalty, by essentially creating an interaction of each pair of basis functions for each marginal term. The basis dimension of the tensor product smooth is the product of the dimensions of the marginal bases. Ex: A tensor product smooth output with 3 effective degrees of freedom means that the model simplified to $\beta_0 + \beta_1 W_i + \beta_2 Z_i + \beta_3 W_i \cdot Z_i$.

Bayesian/‘Random model’ view of smoothing The smoothing penalties on model coefficients can be given a Bayesian interpretation: namely, we have a Gaussian prior on how smooth the function is, and update it after seeing the data. Defining such a prior on model wiggleness actually gives the model the structure of a mixed-effects model. See [Wood \(2017, section 5.8\)](#) for details.

⁵²The effective degrees of freedom tell us how ‘wiggly’ the fitted line is, i.e., the complexity of the estimated smooth function. A value of 1 means that the predictor was estimated to have a linear relationship to the outcome.

⁵³In the R package `mgcv` ::, choosing the argument k (the basis dimension) in spline functions amounts to setting the maximum possible degrees of freedom allowed for each model term. The actual effective degrees of freedom for each term will usually be estimated from the data, by the chosen smoothness selection criterion, but the upper limit on this estimate is $k - 1$: the basis dimension minus one degree of freedom due to the identifiability constraint on each smooth term.

- prior: $f()$ is more likely to be smooth than wiggly: $f(\beta) \propto \exp\left(-\frac{\lambda}{\sigma^2}\beta'S\beta\right) \iff \beta \sim \mathcal{N}\left(0, \frac{\sigma^2}{\lambda}S^{-1}\right)$
- posterior: $\beta|y \sim \mathcal{N}\left(\hat{\beta}, \sigma^2(X'X + \lambda S)^{-1}\right)$ and $\hat{\beta}_{\text{MAP}} = (X'X + \lambda S)^{-1}X'Y$

Estimation We can make use of this parallel between the smoothing penalty on model coefficients and a random effect modeling of these coefficients in two ways: (i) to incorporate random effects into a GAM by treating them as wiggleness penalties, and estimating them without needing to use random effects methods; (ii) or to fit a GAM as a Generalized Additive *Mixed* Model (GAMM).

- **GAM with explicit penalties**

Once a basis and a wiggleness penalty have been chosen for each smooth function, the GAM can be estimated. Estimation is by penalized versions of the least squares⁵⁴ and maximum-likelihood methods used for linear models. The fitting of a GAM has two components:

- (1) Estimating β given λ (estimating the model coefficients under the penalty)
- (2) Estimating the degrees of penalization λ . Multiple methods are possible: prediction error methods, such as Generalized Cross Validation (GCV), or marginal likelihood methods based on the Bayesian/random model view of smoothing.

As aforementioned, this framework also enables to incorporate conventional random effects into the GAM, by representing them like the smooths are represented: as penalized regression terms.⁵⁵

- **GAMM**

All the smooths are converted into random components, in particular the smoothing parameters become components of the variances of the random effect, and the whole model is estimated as a general mixed model (by ML, REML, or PQL).⁵⁶

For example, for a univariate spline $s(X) = \sum_j \beta_j b_j(X)$, the coefficients are represented as random effects, i.e., as belonging to a group: $\beta_j \sim \mathcal{N}(0, \sigma_\beta^2 H)$, where H contains the penalties. At one extreme, an unpenalized likelihood ($\beta_1, \dots, \beta_k$ estimated as fixed effects) would overfit and provide “wiggly” estimates, whereas at the other extreme, a linear fit ($\beta_1 = \dots = \beta_k = 0$) would smooth excessively and capture the relationship with only two parameters.

An advantage of this GAMM approach is that it allows to incorporate correlated error structures, by dealing with them via random effects.

Fezzi and Bateman (2015) uses this method to obtain the optimal (i.e., best linear unbiased predictor) trade-off between excessive smoothing and overfitting of the nonlinear function.

⁵⁴Least squares can be used only in the case of a generalized additive model.

⁵⁵The R function `mgcv::gam` tackles fitting a GAM as a penalized likelihood maximization problem, and uses a separate criterion to estimate the smoothing parameters. The computational strategy is as follows: Basis functions and quadratic penalty matrices are constructed for each smooth term and are combined with a matrix for the strictly parametric part of the model, to form a complete “design” matrix and a set of penalty matrices for the smooth terms. The linear identifiability constraints are also added. The penalized likelihood maximization problem is then solved by Penalized Iteratively Re-weighted Least Squares (PIRLS), and the smoothing parameter estimation problem is solved using the GCV criterion or a Laplace approximation to REML, and is conducted by “outer iteration”: The PIRLS scheme is iterated to convergence for each trial set of smoothing parameters, and GCV or REML criteria are only evaluated on convergence (optimization is ‘outer’ to the PIRLS loop). Random effects can be incorporated by specifying the basis as such. Ex: `s(g, bs='re')` generates an i.i.d. normal random effect; if g is a factor, it produces a random coefficient for each level of g , with the set of coefficients modeled as i.i.d. normal.

⁵⁶In the *normal errors, identity link* case, estimation can be performed using general linear mixed effects modelling software (e.g., in R, as provided by the function `nlme::lme`). This is what the R function `mgcv::gamm` does. It shows it explicitly by returning two items: (i) the fitted model returned by `nlme::lme`, and (ii) an object of class ‘gam’, including a posterior covariance matrix for the parameters of all the fixed effects and the smooth terms.

A.4 Multilevel models

Context We are interested in the relationship of Y_i with X_i . Our data present some hierarchical structure: individuals belong to groups.⁵⁷ Hence observations are not independent, we may expect a group effect $a_{j[i]}$ on the outcome, and we may expect also both a *within*-group relationship and a *between*-group relationship of X_i with the outcome.

Structure A multilevel model is a generalization of a classical regression model that not only allows for variation in coefficients but *models* some of that variation. The generalization can proceed along multiple dimensions, notably:

- (i) whether only intercepts or also slopes are modeled;
- (ii) whether group-level predictors are included.

Let us consider a two-level regression model. The lower level is a common regression model with individual-level predictors. We may assume different distributions for the outcome variable depending on the nature of the data, e.g., normal for continuous data, logistic for binary data, negative binomial for count data... Now some of its coefficients (the intercept and/or slopes) are allowed to vary by group and are *modeled*, i.e., are given a probability model, which constitutes the higher- or group-level model.

The generalization along the aforementioned dimensions gives rise to model types like the below:⁵⁸

$\forall i = 1, \dots, n, \quad j = 1, \dots, J:$

	w/o group-level predictors	with group-level predictors
varying intercept	$Y_i \sim \mathcal{D}_1(a_{j[i]} + \beta X_i, \sigma_Y^2)$ e.g. $Y_i^{\text{real}} \sim \mathcal{N}(a_{j[i]} + \beta X_i, \sigma_Y^2)$ or $Y_i^{\text{binary}} \sim \text{Ber}(\exp(a_{j[i]} + \beta X_i))$ or $Y_i^{\text{count}} \sim \text{NB}(\exp(a_{j[i]} + \beta X_i), r)$ $a_j \sim \mathcal{D}_2(\mu_a, \sigma_a^2)$	$Y_i \sim \mathcal{D}_1(a_{j[i]} + \beta X_i, \sigma_Y^2)$ $a_j \sim \mathcal{D}_2(\alpha_0 + \alpha_1 Z_j, \sigma_a^2)$
varying intercept and slope	$Y_i \sim \mathcal{D}_1(a_{j[i]} + b_{j[i]} X_i, \sigma_Y^2)$ $a_j \sim \mathcal{D}_2(\mu_a, \sigma_a^2)$ $b_j \sim \mathcal{D}_3(\mu_b, \sigma_b^2)$	$Y_i \sim \mathcal{D}_1(a_{j[i]} + b_{j[i]} X_i, \sigma_Y^2)$ $a_j \sim \mathcal{D}_2(\alpha_0 + \alpha_1 Z_j, \sigma_a^2)$ $b_j \sim \mathcal{D}_3(\beta_0 + \beta_1 L_j, \sigma_b^2)$
co-varying intercept and slope	$Y_i \sim \mathcal{D}_1(a_{j[i]} + b_{j[i]} X_i, \sigma_Y^2)$ $\begin{bmatrix} a_j \\ b_j \end{bmatrix} \sim \mathcal{D}_{ab} \left(\begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{bmatrix} \right)$	$Y_i \sim \mathcal{D}_1(a_{j[i]} + b_{j[i]} X_i, \sigma_Y^2)$ $\begin{bmatrix} a_j \\ b_j \end{bmatrix} \sim \mathcal{D}_{ab} \left(\begin{bmatrix} \alpha_0 + \alpha_1 Z_j \\ \beta_0 + \beta_1 L_j \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{bmatrix} \right)$

Terminology and relation to classical models

Let us see what assumptions these models embed, and how these compare to those of classical single-level models. For simplicity, we consider a normally-distributed outcome and a single regressor X_i with a homogeneous effect β , i.e., group effects manifest only through the intercept).

⁵⁷For example, we have observations of n students in J classrooms such that each Student's i belongs to a group $j[i]$, or we have longitudinal data, where each dated observation $\{i, t\}$ belongs to a unit i . For simplicity, we consider in this section only such hierarchical data with two levels. However, results extend to more levels, that can be nested or not.

⁵⁸For clarity, we use Latin letters for variables and modeled parameters, and Greek letters for fixed parameters (as distinguished in a frequentist framework). The fixed parameters that inform varying parameters are called the “hyperparameters” of the full model.

Any model fit embeds some assumption about how these group effects $\{a_{j[i]}\}_j$ are distributed. And notably about their variance σ_a^2 , which shows how close they are to each other, i.e., how much they are pooled toward their mean. Classical models correspond to extreme assumptions about this variance, while a multilevel model is a data-driven compromise:

- *Complete pooling*: We don't allow for group effects.
 - Conceptualization: No group effects.
 - Implementation: We neither model them nor adjust for them. We fit the basic model:

$$\begin{aligned} Y_i &= \alpha + \beta X_i + e_i, & e_i &\sim \mathcal{N}(0, \sigma_Y^2) \\ \iff Y_i &= a_{j[i]} + \beta X_i + e_i, & e_i &\sim \mathcal{N}(0, \sigma_Y^2), \quad a_j \sim \mathcal{D}(\alpha, 0) \end{aligned} \quad (\text{m1})$$

- *No pooling*: We allow for group effects but don't model them.
 - Conceptualization: The group effects are “fixed effects”, unrelated to each other.
 - Implementation: We fit a separate intercept per group (by including a dummy variable for each j) or de-mean the data by group, which eliminate the group effects. We fit the “fixed effects” model:

$$\begin{aligned} Y_i &= \alpha_{j[i]} + \beta X_i + e_i, & e_i &\sim \mathcal{N}(0, \sigma_Y^2) \\ \iff Y_i &= a_{j[i]} + \beta X_i + e_i, & e_i &\sim \mathcal{N}(0, \sigma_Y^2), \quad a_j \sim \mathcal{D}(\alpha, \infty) \\ \iff (Y_i - \bar{Y}_{j[i]}) &= \beta (X_i - \bar{X}_{j[i]}) + e_i, & e_i &\sim \mathcal{N}(0, \sigma_Y^2) \end{aligned} \quad (\text{m2})$$

- *Partial pooling*: We allow for group effects and model them.
 - Conceptualization: The group effects are “random effects” with a joint probability distribution.
 - Implementation: We model them with error, i.e., we estimate the parameters of that distribution. The $\{a_j\}$ s are pooled toward their common mean μ_a , by an amount that depends on the sample size of each group and on σ_a^2 ,⁵⁹ which is also estimated from the data. We fit the multilevel or “random effects” model:⁶⁰

$$\begin{cases} Y_i = a_{j[i]} + \beta X_i + e_i & e_i \sim \mathcal{N}(0, \sigma_Y^2) \\ a_j \sim \mathcal{D}(\mu_a, \sigma_a^2) \end{cases} \quad (\text{m3})$$

Partial pooling is thereby a data-driven compromise between *complete pooling* (equivalent to setting $\sigma_a \rightarrow 0$) and *no pooling* (setting $\sigma_a \rightarrow \infty$). The former assumes away variation between groups, while the latter risks overstating that variation, i.e., overfitting the data, and neither lets us analyze it (Gelman and Hill, 2006, ch. 12).

Why use multilevel models?

In short: to (1) acknowledge and (2) analyze within-group *and between-group* variations.

- *To account for the group-dependence in our data (e.g., with time series, spatial correlation, networks...)*. Traditional regression techniques assume independent observations. Any dependence structure in our data that isn't modeled will be left out in the error term, and the corresponding standard errors of regression coefficients will be underestimated (esp. that of higher-level regressors). A common way to deal with group-dependence in econometrics is to fit group dummies and to *cluster standard errors* by group (after fitting the model, a new estimator of the error covariance matrix is computed that adjusts for the dependence within groups). Instead, a multilevel model models this group-dependence. It is

⁵⁹Partial pooling is proportional to the variance, not the standard deviation.

⁶⁰In the regression framework, multilevel regression models are a particular case of “random effect” models that pool information across groups, or “mixed effect” models when they include both a “random effect” component and a “fixed effect” component (e.g., a varying-intercept, fixed-slope model). The “fixed effect” vs “random effect” terminology is confusing, as different disciplines use these to refer to different things. For instance, in econometrics, a fixed effect is an intercept that varies by group (and that is estimated using dummy variables, i.e., that isn't modeled), while in other branches of statistics, it means a coefficient that does not vary. Gelman and Hill (2006, p. 245) suggests to avoid these terms and instead describe models explicitly: for example, a ‘varying intercepts and constant slopes’ model.

equivalent to a classical regression model with correlated *and modeled* errors:⁶¹

$$\begin{cases} Y_i = \alpha_{j[i]} + X_i' \beta + e_i, & e_i \sim \mathcal{N}(0, \sigma_Y^2) \\ \alpha_j = \mu_\alpha + \eta_j, & \eta_j \sim \mathcal{N}(0, \sigma_\alpha^2) \end{cases} \iff Y_i = \mu_\alpha + X_i' \beta + \underbrace{e_i + \eta_{j[i]}}_{e_i^{\text{all}}}, \quad e^{\text{all}} \sim \mathcal{N}(0, \Sigma)$$

- *To increase efficiency by pooling information.* By treating groups as a “random-effect” within the model, we can pool shared information about the mean across the groups. *Partially pooling* the varying coefficients will produce more efficient (less noisy) estimates of the J regression lines than by including group indicators, especially when the number of observations in some groups is small.⁶²
- *To model heterogeneity in the relationship to a covariate.* For example, in causal inference, we may be interested not just in an average treatment effect but also in how the effect varies across the population. With a multilevel model, we can model variation in the expected treatment effect, for example as a function of pre-treatment covariates x .
- *To generalize results to a population not well-represented in the sample.*
 - To do inference for the population of groups when our data are not random samples: one can generalize to a larger population using *multilevel regression and poststratification (MRP)*.
 - To estimate $\widehat{Y}$ for particular groups: notably to get reasonable estimates even for groups with small sample sizes (which is difficult with classical regression).
 - To predict a new observation in a new group: multilevel regression enables to quantify sources of variation, and hence to propagate the uncertainty about the new group into the uncertainty about the new individual in this group; this distinction isn’t provided in classical regression.

Ex: With fixed effects, these group effects are allowed to take any value whatsoever, which amounts to saying that each individual group is completely different to every other group: having results for 6 groups tells us nothing about a 7th. Treating the groups instead as a random sample from the population of groups allows us to estimate the relationship between Y_i and X_i and to generalize beyond the 6 groups in the sample. Each new group would indeed have the same distribution as that of the 6 groups in the sample.

⚠ Endogeneity concern in causal inference, mitigated by REWB

Context: We want to estimate the causal effect β of X_i on Y_i . But there are unobserved group-level or “between” effects that are correlated with X_i and determine Y_i . Hence the slope estimator of the “within effect” in a simple regression of Y_i on X_i would suffer from omitted variable bias.⁶³ How to tackle this bias on level 1 coefficients due to omitted variables at level 2?

We saw that the J unobserved group effects can be considered as either *fixed effects* (i.e., unrelated) or *random effects* (i.e., belonging to a shared distribution). The FE approach provides an unbiased estimator of the *within* effect, with the caveats that (i) the assumption of unrelatedness is very questionable, and (ii) no group-level variable can be identified as all the group-level variance is accounted for. However, the simple RE model presents another important problem: its estimator $\hat{\beta}$ is biased for the within effect. Indeed, a correlation with $\bar{X}_i$ is left in the error term, as can be seen by rewriting the model as below, where η_j contains these problematic unobserved group-level effects that are correlated with X_i and determine Y_i :

$$\begin{cases} Y_i = a_{j[i]} + X_i' \beta + e_i, & e_i \sim \mathcal{N}(0, \sigma_Y^2) \\ a_j = \mu_\alpha + \eta_j, & \eta_j \sim \mathcal{N}(0, \sigma_a^2) \end{cases} \iff Y_i = \mu_\alpha + X_i' \beta + \underbrace{e_i + \eta_{j[i]}}_{e_i^{\text{all}}}, \quad e^{\text{all}} \sim \mathcal{N}(0, \Sigma)$$

⁶¹The error e_i^{all} is the sum of an individual-level noise e_i and a group-level error $\eta_{j[i]}$ which induces correlation in e^{all} . The covariance matrix Σ is parameterized in some way, and these parameters are estimated from the data.

⁶²Note however that when the number of groups J is small, it is difficult to estimate the between-group variation σ_a^2 precisely. As this σ_a^2 determines the amount of partial pooling, a bad estimation of its value results in pooling by a somewhat random amount. Hence multilevel modeling adds little to no-pooling models.

⁶³This endogeneity concern can arise with any form of nested data. A very common case is that of longitudinal data (it = level 1, i = level 2), where the concern is of time-constant effects that are plausibly correlated with X_{it} .

The bias can be described: the estimator $\hat{\beta}_{\text{RE}}$ is actually a weighted average of the within estimator and the between estimator.⁶⁴

A solution is to add $\bar{X}_{j[i]}$ as a regressor, as it will extract the problematic correlation from the composite error term e_i^{all} . We recover an estimator of the within effect that is not biased by group-level confounders.⁶⁵ The model is known as the ‘within-between RE’ (REWB) model (Bell et al., 2019), as it decomposes X into a within-group component and a between-group component, or ‘correlated RE’ (CRE) model (Wooldridge, 2013, 14.3). The three expressions of the model below are equivalent:⁶⁶

$$\begin{cases} Y_i = a_j + \beta_1(X_i - \bar{X}_j) + \beta_2\bar{X}_j + e_i \\ a_j \sim \mathcal{N}(\mu_a, \sigma_a^2) \end{cases} \iff \begin{cases} Y_i = a_j + \beta_1(X_i - \bar{X}_j) + e_i \\ a_j \sim \mathcal{N}(\alpha + \beta_2\bar{X}_j, \sigma_a^2) \end{cases} \iff \begin{cases} Y_i = \beta_0 + \beta_1(X_i - \bar{X}_j) + \beta_2\bar{X}_j + \nu_j + e_i \\ \nu_j \sim \mathcal{N}(0, \sigma_\nu^2) \text{ uncorrelated w. } X_i \end{cases}$$

The REWB model also makes it possible to:

- estimate the between effect, as well as any relationship between the outcome and a group-level covariate (though one couldn’t interpret its coefficient as causal);
- test whether the between and the within effects are significantly different (with a Hausman test of whether the difference between the two coefficients in the model is statistically different from 0);
- estimate the level-2 variance and compare it to the level-1 variance.

The REWB model can also naturally be extended to varying intercepts *and slopes* and/or additional group-level predictors.⁶⁷

CCL: We should use FEs only when we really don’t care about between-group relationships + don’t worry about efficiency (as e.g. n is very large).

Inference These models either can be estimated under either the frequentist or the Bayesian paradigms, and using various computational methods (deterministic vs simulation-based) from various packages.⁶⁸ For example, in R:

- **with lme4::** (*frequentist; deterministic*)

First the hyperparameters are estimated via Maximum Likelihood or Restricted Maximum Likelihood (REML), then inference is performed for the coefficients conditional on the estimated hyperparameters. We obtain point estimates and asymptotic standard errors based on asymptotic likelihood theory.

⚠ ML estimators are unbiased for the group-level mean parameters but downward-biased for the group-level variance parameters (especially when the number of groups is small), because the mean

⁶⁴ $\beta_1 = \frac{w_W\beta_W + w_B\beta_B}{w_W + w_B}$, where $w_W := 1/\text{SE}[\hat{\beta}_W]^2$ and $w_B := 1/\text{SE}[\hat{\beta}_B]^2$ are the precisions of the within estimate and the between estimate, respectively. As there are more data at level 1 (and therefore higher precision of the within estimate), $\hat{\beta}_1$ will often tend towards the within estimate (Bell et al., 2019).

⁶⁵⚠ With a *non-identity link function*, unbiasedness is guaranteed only if u_j is a linear function of $\bar{X}_{j[i]}$. I.e., to get an unbiased effect, we are trading one assumption of linearity for another (in the standard model, we assume an identity link function; in the REWB model with non-identity link function, we assume u_j is a linear function of $\bar{X}_j$). However, the available evidence (from simulations) suggests that the bias of the REWB method remains small in most situations (Bell et al., 2019). One can also include functions of $\bar{X}_{j[i]}$ as regressors to characterize more flexible functional forms of the correlation. P. Allison suggests using polynomial functions of the means, i.e., including not only $\bar{X}_i$ but also $\bar{X}_i^2, \bar{X}_i^3$ as regressors, or other cluster-level functions of the X_{it} , such as their standard deviation (see <https://statisticalhorizons.com/problems-with-the-hybrid-method/>). If the estimates of the added coefficients aren’t significant and the estimate of β_W doesn’t change much, it suggests the linearity assumption is reasonable and bias should not be such an issue.

⁶⁶We center level-1 variables, i.e., we use as regressor $(X_{it} - \bar{X}_{j[i]})$ rather than X_i , such that there is no correlation between $(X_{it} - \bar{X}_{j[i]})$ and $\bar{X}_{j[i]}$, and β_2 represents the *between* effect rather than the *contextual* effect (Bell et al., 2019).

⁶⁷For example:
$$\begin{cases} Y_i = a_j + b_j(X_i - \bar{X}_j) + e_i \\ \begin{bmatrix} a_j \\ b_j \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} \beta_0 + \beta_2\bar{X}_j \\ \beta_1 \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{bmatrix} \right) \end{cases} \iff Y_i = \beta_0 + \beta_1(X_i - \bar{X}_j) + \beta_2\bar{X}_j + \nu_{i0} + \nu_{i1}(X_i - \bar{X}_j) + e_i,$$

where the composite error term $\nu_{i0} + e_i$ is uncorrelated with X_i . For unbiasedness, the group average need not be included as a predictor for the slope?

⁶⁸⚠ The computational method used for estimation (deterministic vs simulation-based) need not be determined by the statistical paradigm (frequentist vs Bayesian, i.e., how parameters and probability are interpreted). R’s **lme4::** does not use simulations simply because the ML estimator can be solved for using deterministic optimization algorithms and approximations, whereas R’s **rstanarm::** does use simulations because the posterior $p(\theta|Y)$ of a hierarchical regression model is too complicated to derive analytically; Stan therefore samples from it using HMC/NUTS. However, in other cases, e.g. when models have closed-form posteriors, Bayesian estimation can be pursued deterministically.

parameters are assumed to be known with certainty when estimating the variance parameters. Instead, REML accounts for the number of mean parameters estimated, losing 1 degree of freedom for each, and so produces unbiased estimators of variance parameters. Note however that to compare models with *different fixed effects* with a likelihood ratio test, ML must be used, as LR tests for REML require exactly the same fixed effects specification in both models.

👍 Computational: fast.

🗣️ A. Gelman: “*The usual non-Bayesian procedures are designed to work well asymptotically (in the case of hierarchical models, this is the limit as the number of groups approaches infinity). But as noted Bayesian J. M. Keynes could’ve said, asymptotically we’re all dead.*”

- **with `rstanarm::`**—or with `rstan::stan()` and coding the model in the Stan language—(*Bayesian; simulation-based*)

All levels are fitted simultaneously. The hyperparameters are given a prior distribution, and we estimate their whole posterior distribution. We obtain posterior samples, i.e., simulated draws of the posterior distributions, and summaries.

👍 Accounts for all the uncertainty in the parameter estimates when predicting the varying intercepts and slopes, and their associated uncertainty.

🗣️ Computational: slow. Markov chain Monte Carlo simulations are generally much slower than (RE)ML estimation.

B Reading regression output

B.1 Diagnostics of modeling assumptions

After our statistical software has fit the model and produced the estimates requested, we should check that the assumptions underlying these numbers hold. Several key assumptions can be diagnosed by simply looking at the residuals, which contain the variation in Y and the features of the relationship between Y and X that weren't explained by the model.

Let us consider the simplest example of the estimation of a CLRM. The four plots presented below provide information w.r.t. a specific assumption of the CLRM.⁶⁹ In each figure, the “Case 1” and “Case 2” subplots illustrate cases where the given assumption seems to be met relatively well, vs the reverse, respectively.

1. “Residuals vs Fitted” plot — *Is there an unmodeled non-linear pattern?*

Residuals are plotted against fitted values. Are the residuals spread rather equally around a horizontal line, without distinct patterns? If they aren't, it suggests that a non-linear relationship was not explained by the model and was therefore left out in the residuals. *Note: If Y_i is discrete, such as in a logistic regression, then residuals are discrete. One shouldn't plot raw residuals, but rather binned residuals (divide the data equally into bins based on fitted values, s.t. each bin has the same number of points, and take the averages for each bin).*

2. “Scale-Location” plot — *Are the residuals homoscedastic?*

The square root of the absolute value of standardized residuals $\sqrt{|r_i|}$ is plotted against fitted values $\hat{Y}_i$. Is the vertical spread of points uniform along X ? A uniform spread indicates residuals have a uniform variance across the range of predicted values. The reverse suggests there is heteroscedasticity.

3. “Normal Q-Q” plot — *Are the residuals normally distributed?*

The quantiles of the residuals are plotted against the theoretical quantiles of the normal distribution. If the residuals are approximately normally distributed, we should see a roughly straight line. The plot may also reveal outliers.

4. “Residuals vs Leverage” plot — *Are there influential observations?*

First, let's distinguish outliers, high leverage points, and influential points:

- Outliers are observations with unusual outcome values Y_i (i.e., that are considerably different from the rest of the data). They may not have a lot of influence on the regression line.
- High-leverage points are observations with unusual predictor values x_i . In linear regression, leverage measures how sensitive a fitted $\hat{Y}_i$ is to a change in the true Y_i . High-leverage points will not have a lot of influence on the regression line if they lie close to it.
- Finally, influential points are observations whose removal from the data would cause a large change in the estimated regression line. I.e., they largely disagree with the trend.

A point has to have at least some leverage in order to be influential. To identify influential points, we can compute each observation's Cook's distance d_i , which measures the effect of omitting that observation on the parameter vector. Precisely, d_i has a component that reflects how well the model fits the i -th observation Y_i and a component that measures how far that point is from the rest of the data. Points with $d_i > 1$ are generally considered influential.

In the plot, residuals are plotted against their leverage, and dotted red lines represent Cook's distances of 0.5 and 1. Points outside these lines have high Cook's distances, i.e., high influence.⁷⁰

⁶⁹These four specific types of plots are built-in diagnostic plots for linear regression analysis in R (one need simply run `plot()` on the fitted model object). The figures used here for illustrative purposes are taken from <https://data.library.virginia.edu/diagnostic-plots/>.

⁷⁰If the lines aren't visible on the graph, it means that all points are well inside them: there are no influential points.

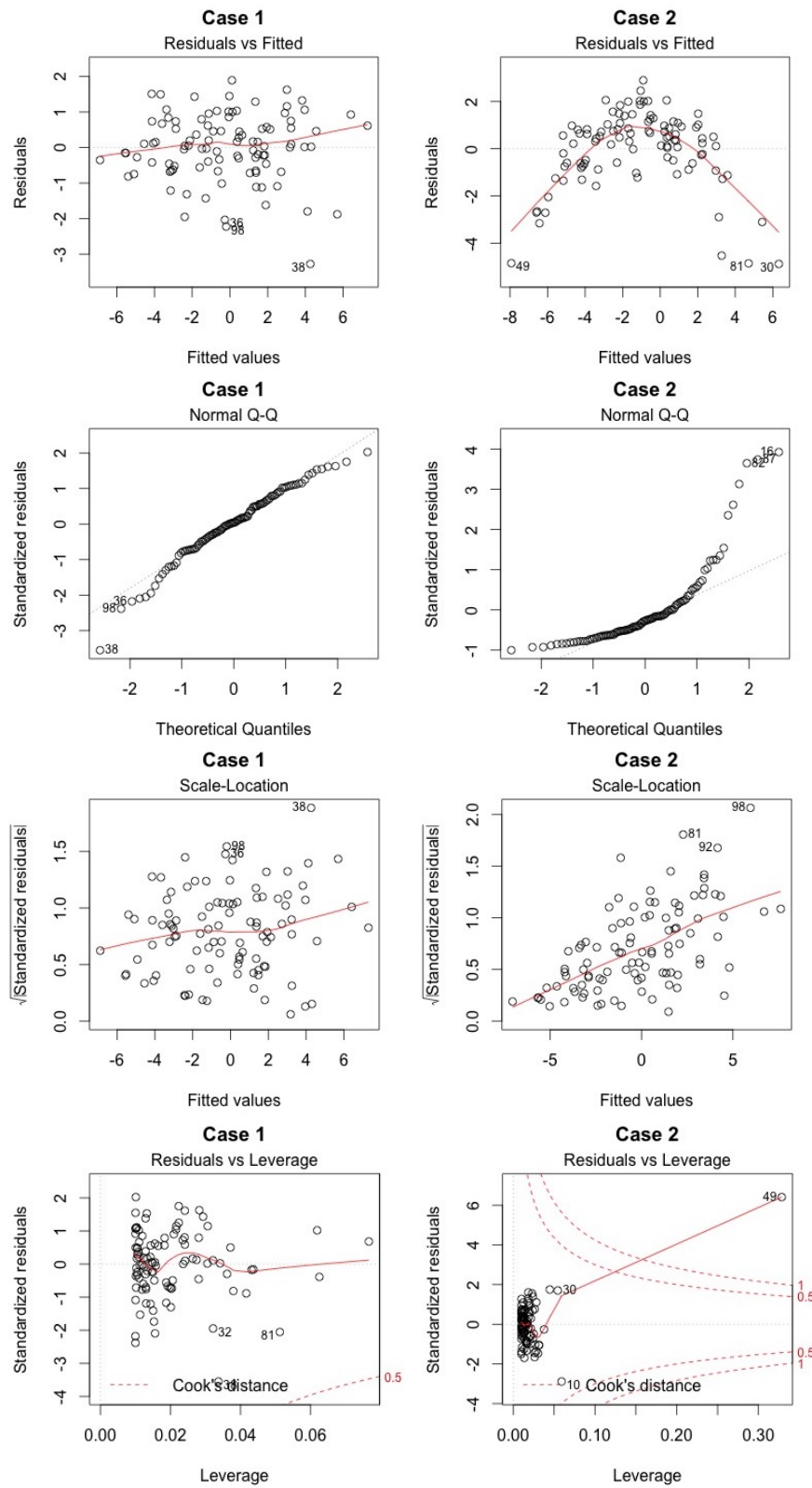


Figure 1: Diagnostic plots. From top to bottom: Residuals vs Fitted; Scale-Location; Normal Q-Q; Residuals vs Leverage.

B.2 Linear model estimated by OLS

```
> mod1 = lm(Ozone ~ Wind + Temp + Wind:Temp, data = airquality)
> summary(mod1)

Call:
lm(formula = Ozone ~ Wind + Temp + Wind:Temp, data = airquality)

Residuals:
    Min       1Q   Median       3Q      Max
-39.906 -13.048  -2.263   8.726  99.306

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -248.51530    48.14038  -5.162 1.07e-06 ***
Wind          14.33503     4.23874   3.382 0.000992 ***
Temp           4.07575     0.58754   6.937 2.73e-10 ***
Wind:Temp     -0.22391     0.05399  -4.147 6.57e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.44 on 112 degrees of freedom
(37 observations deleted due to missingness)
Multiple R-squared:  0.6261,    Adjusted R-squared:  0.6161
F-statistic: 62.52 on 3 and 112 DF,  p-value: < 2.2e-16
```

Figure 2: Summary of the results of the OLS estimation of a multivariate linear regression model $Y_i = X_i' \beta + e_i$, $e_i \stackrel{\text{iid}}{\sim} (0, \sigma^2)$, $i = 1, \dots, 116$, in R.

Residuals $\{r_i\}_i := Y_i - \hat{Y}_i = Y_i - X_i' \hat{\beta}$ are estimates of the errors $u_i := Y_i - X_i' \beta$. One should plot them to check that their distribution matches our model assumptions; e.g., whether is it centered around 0, normal...

Coefficients For each coefficient, a point value estimate, a level of uncertainty for that estimate, and a two-sided t-test for the null hypothesis of a zero value.

- **Slope estimate** $\hat{\beta}_j^{\text{OLS}} := \arg \min_{\beta} \sum_{i=1}^n r_i^2$

Interpretation is contingent on the other regressors $X_{l \neq j}$ in the model. Indeed, each slope estimand is $\beta_j = \frac{\text{cov}[\tilde{X}_j, Y]}{\text{v}[\tilde{X}_j]}$ where $\tilde{X}_j$ is the residual from the regression of X_j on all the other covariates $X_{l \neq j}$.⁷¹

Interpretation is furthermore as *comparison between units*⁷² that differ in X_j . Hence:

- For a continuous X_j : “ $\hat{\beta}_j$ is the average difference in Y for a 1-unit difference in X_j across units, holding other regressors constant;”
- For a categorical X_j : “ $\hat{\beta}_j$ is the average difference in Y between a unit in the given category and one in the reference category, holding other regressors constant;”
- With interactions: the slope of one variable is allowed to vary across the support of the other, so the interpretation of coefficients varies. Ex: In the example above, the difference in Y associated with $Wind$ is $\beta_1 + \beta_3 Temp$: it is different for each value of $Temp$.

- **Standard Error** $\text{SE}[\hat{\beta}_j] := \sqrt{\text{Var}(\hat{\beta}_j)}$ is an estimate of the standard deviation of $\hat{\beta}_j$'s sampling distribution. It can be thought of as the expected difference in $\hat{\beta}_j$ if we were to run the model again and again. The lower it is *relative to the estimated point value*, the higher our uncertainty about $\hat{\beta}_j$.

⁷¹In the univariate regression model it simplifies to $\beta = \frac{\text{cov}[X, Y]}{\text{v}[X]}$, with the estimator its sample analog $\hat{\beta} = \frac{\frac{1}{n} \sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{n} \sum_i (X_i - \bar{X})^2}$.

⁷²The estimates of $\hat{\beta}$ should be interpreted as “effects” only in causal inference. A causal identification strategy is required for a “counterfactual” interpretation as *the expected change in Y caused by adding 1 to X* , i.e., as a change *within* units. Otherwise, a regression only tells us about comparisons between units, not about changes within units.

- **t-statistic** $t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - 0}{\text{SE}[\hat{\beta}_j]}$ is the realization of the t -statistic for the null hypothesis $\{H_0 : \beta_j = 0\}$.

If $\hat{\beta}_j$ is normally distributed, then $t_{\hat{\beta}_j} \underset{H_0}{\sim} \mathcal{T}_{n-p}$, which can be used in a t -test of H_0 .

- **two-sided⁷³ p-value** $p\text{-value}_{\hat{\beta}_j} = \Pr(\text{observing a } |T| \geq |t_{\hat{\beta}_j}| \text{ under } H_0)$ is the probability of observing data as extreme as that actually observed, assuming H_0 .

$p\text{-value}_{\hat{\beta}_j}$ small (< 0.05) $\iff t_{\hat{\beta}_j}$ falls in the tails of the Student's $\mathcal{T}$ -distribution
 $\implies$ observing our $t_{\hat{\beta}_j}$ is highly unlikely under H_0
 $\implies$ reject H_0
 $\implies$ there is a relationship between Y and X_j , $\hat{\beta}_j$ is “significant.”

Residual Standard Error Deviation $= \sqrt{\frac{\text{RSS}}{\text{df}_{\text{RSS}}}} = \sqrt{\frac{\sum_i r_i^2}{n-p}}$

It's the standard deviation of the residuals, i.e., an estimate of the standard deviation of the errors. Its square is the variance in Y that is unexplained by the model. It captures the accuracy to which the model can predict Y , as the average distance of an observed value to the regression line: “ Y can deviate from the regression line by 20.44 ppb, on average”; “The model predicts Y to an accuracy of about 20.44 points.”

R^2 $= 1 - \frac{\text{RSS}}{\text{RSS}_{\text{null}}} = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{ESS}}{\text{TSS}} \in [0, 1]$ is the share of the variance in Y that is explained by X ,⁷⁴ where we define:

- The *null* model as the model with no predictors. It implies $\forall i, \hat{Y}_i = \bar{Y}$.
 - The model's *residual sum of squares* $\text{RSS} := \sum_i (Y_i - \hat{Y}_i)^2 = \sum_i r_i^2$ (aka sum of squares of errors – SSE);
 - The model's *explained sum of squares* $\text{ESS} := \sum_i (\hat{Y}_i - \bar{Y})^2$ (aka sum of squares due to regression – SSR);
 - The model's *total sum of squares* $\text{TSS} := \sum_i (Y_i - \bar{Y})^2$.
- The null model has $\forall i, \hat{Y}_i = \bar{Y}$, hence $\text{RSS}_{\text{null}} = \sum_i (Y_i - \bar{Y})^2 = \text{TSS}$. As $\text{TSS} = \text{ESS} + \text{RSS}$, $R^2 = \dots = \frac{\text{ESS}}{\text{TSS}}$ = the proportion of the sample variance in Y that is explained by X .

△ R^2 mechanically increases as more predictors are included in the regression. → One should report the **adjusted R^2** which adjusts for the number of predictors.

△ One should never select a model based on R^2 , as this would favor overfitting.

F statistic for an F-test⁷⁵ of **overall significance** of the model. *Is our model better than a model with no regressors? I.e., are the $p-1$ slope coefficients jointly significant?*

We test the full model against the null model with no regressors. $H_0 : \beta_j = 0, \forall j$

$$F = \frac{\text{ESS}/\text{df}_{\text{ESS}}}{\text{RSS}/\text{df}_{\text{RSS}}} = \frac{\text{ESS}/(p-1)}{\text{RSS}/(n-p)} = \frac{\text{mean squares due to regression (MSR)}}{\text{mean squares of errors (MSE)}} \in [0, +\infty[$$

If $\hat{\beta}$ is normally distributed, then $F \underset{H_0}{\sim} \mathcal{D}_{p-1, n-p}$. We can hence use the realized F statistic to conduct an F -test of H_0 . If F 's p-value is small, we may conclude that at least some of the parameters are non-zero and the regression equation has some validity in fitting the data.

⁷³By default, most statistical packages carry out a two-sided test and hence report the two-sided p-value; however, we could also use the t-statistic to carry out a one-sided test.

⁷⁴Mathematically, the sum of squared deviations is an unscaled measure of dispersion or “variability”. When scaled for the number of degrees of freedom, it estimates the variance, i.e., the spread of the data about their mean value. Partitioning of the sum of squared deviations into various components allows the overall variability in a dataset to be ascribed to different sources.

⁷⁵In general, an F -statistic is a ratio of two quantities, and is used in an F -test of the H_0 that the quantities are roughly equal, i.e. $F \approx 1$. One rejects H_0 if F is high (> 1). How large F really needs to be depends on the number of data points and predictors: if large sample, F -stat slightly above 1 is sufficient to reject H_0 ; if small sample, need a large F -stat.

B.3 Generalized linear model estimated by ML

```

> titanic <- as.data.frame(Titanic) %>%
+   tidyr::uncount(Freq) %>% mutate(Survived = 1*(Survived=="Yes"))
> mod2 = glm(Survived ~ Sex + Age + Class, data = titanic,
+            family = binomial(link = 'logit'))
> summary(mod2)

Call:
glm(formula = Survived ~ Sex + Age + Class, family = binomial(link = "logit"),
    data = titanic)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.6853      0.2730   2.510   0.0121 *
SexFemale     2.4201      0.1404  17.236 < 2e-16 ***
AgeAdult     -1.0615      0.2440  -4.350 1.36e-05 ***
Class2nd      -1.0181      0.1960  -5.194 2.05e-07 ***
Class3rd      -1.7778      0.1716 -10.362 < 2e-16 ***
ClassCrew     -0.8577      0.1573  -5.451 5.00e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2769.5  on 2200  degrees of freedom
Residual deviance: 2210.1  on 2195  degrees of freedom
AIC: 2222.1

Number of Fisher Scoring iterations: 4

```

Figure 3: Summary of the results of the ML estimation of a multivariate logistic regression model $Y_i|X_i \sim \text{Ber}(\pi_i)$, $\text{logit}(\pi_i) = X_i'\beta$, $i = 1, \dots, 2201$, in R.

Coefficients For each coefficient, a point value estimate, a level of uncertainty for that estimate, and a z-test for the null hypothesis of a zero value.

- **Slope estimate** $\hat{\beta}_j^{\text{MLE}} := [\arg \max_{\beta} \mathcal{L}(\mathbf{Y} | \mathbf{X}, \beta)]_{jj}$ is the value of β that maximizes the likelihood function $\mathcal{L}$ (the probability density of the data given the parameters and predictors), i.e., that makes the observed data most likely.
- **Asymptotic Standard Error** $\text{ASE}[\hat{\beta}_j] = \sqrt{[(\mathbf{X}'\widehat{\mathbf{W}}\mathbf{X})^{-1}]_{jj}}$ ML estimators generally don't have an exact finite-sample formula for $\widehat{\mathbf{V}}(\hat{\beta}_j)$, so there is no exact finite-sample SE like there is in OLS. R's `glm()` actually reports an *asymptotic* SE, which stems from $\hat{\beta}$ being asymptotically normally-distributed: $\hat{\beta} \stackrel{a}{\sim} \mathcal{N}(\beta, (\mathbf{X}'\widehat{\mathbf{W}}\mathbf{X})^{-1})$. In logistic regression, $\widehat{\mathbf{W}} := \text{diag}(\hat{\pi}_i(1 - \hat{\pi}_i))$.
- **z-statistic** $z_{\hat{\beta}_j} = \frac{\hat{\beta}_j - 0}{\text{ASE}[\hat{\beta}_j]} \stackrel{a}{\underset{H_0}{\sim}} \mathcal{N}(0, 1)$
- **two-sided p-value** $p_{\hat{\beta}_j} = \Pr(\text{observing a } |Z| \geq |z_{\hat{\beta}_j}| \text{ under } H_0)$

I.e., the probability of observing data as extreme as that actually observed, assuming H_0 .

$p\text{-value}_{\hat{\beta}_j}$ small (< 0.05) $\iff z_{\hat{\beta}_j}$ falls in the tails of the standard normal distribution
 $\implies$ observing our $z_{\hat{\beta}_j}$ is highly unlikely under H_0
 $\implies$ reject H_0
 $\implies$ there is a relationship between y and x_j , $\hat{\beta}_j$ is “significant.”

Deviances are the building blocks to test the **overall significance** of the model. Similarly to OLS, we ask: *Is our model a significantly better fit than a model with no regressors? I.e., are the p slope coefficients jointly significant?*

We want to test the full model against the null model with no regressors. $H_0: \beta_j = 0, \forall j$.

We run a **likelihood-ratio test**:

- The deviance of a model is a log-likelihood ratio between it and the *saturated* model (a model with a parameter for every observation s.t. the data are fitted exactly): $D := 2(\log \mathcal{L}_{\text{sat}} - \log \mathcal{L}_{\text{model}})$.
- The **null deviance** D_{null} is the deviance of the null (no predictors) model;
- The **residual deviance** D is the deviance of our regression model.
- Under H_0 , their difference follows an asymptotic chi-square distribution: $D_{\text{null}} - D \stackrel{a}{\sim}_{H_0} \chi^2_{df[D_{\text{null}}-D]}$. We can hence run a statistical test of H_0 using the realized $D_{\text{null}} - D$.⁷⁶

In our example, $D_{\text{null}} - D = 2769.5 - 2210.1 = 559.4 \stackrel{a}{\sim}_{H_0} \chi^2_{2200-2195=5}$. We find the p-value: $p = \Pr(\chi^2_5 \geq 559.4) = 1.192787e-118$. $p \leq 0.05 \rightarrow$ We reject H_0 . $\rightarrow$ The model is overall significant, it's a significantly better fit than the intercept-only model.

⁷⁶We note the parallel between OLS and MLE: The deviance of a model is the MLE generalization of OLS's RSS, as, for the linear model, maximizing the likelihood is equivalent to minimizing the RSS. In their respective test of the overall significance of the models, each method uses a comparison of the performance of the given regression model to that of the null model, scaled by the degrees of freedom of that comparison. Indeed, $D_{\text{null}} - D = \dots = 2(\log \mathcal{L}_{\text{model}} - \log \mathcal{L}_{\text{null}})$.

B.4 Multilevel models estimated via simulations

This section documents raw output from R's `rstanarm` package, which performs estimation via simulations.⁷⁷

- Varying intercept, no group-level predictors

$$\begin{cases} y_i^{\text{count}} \sim \text{NB}(\exp(a_{j[i]} + \beta x_i), r) \\ a_j \sim \mathcal{N}(\mu_a, \sigma_a^2) \end{cases}$$

```
> m1 = rstanarm::stan_glmer.nb(y_count ~ 1 + x + (1 | group), mydata)
> rstanarm::summary(m1)
```

	mean	mcse	sd
(Intercept)	-0.812	0.004	0.114
x	0.201	0.002	0.101
b[(Intercept) group:1]	0.002	0.007	0.213
...			
b[(Intercept) group:30]	0.028	0.006	0.195
reciprocal_dispersion	1.927	0.031	0.785
Sigma[group:(Intercept),(Intercept)]	0.055	0.004	0.067
mean_PPD	0.473	0.002	0.067
log-posterior	-321.526	0.374	4.911

```
> rstanarm::print(m1)
```

	Median	MAD_SD
(Intercept)	-0.809	0.116
x	0.202	0.095

```
Auxiliary parameter(s):
```

	Median	MAD_SD
reciprocal_dispersion	1.736	0.661

```
Error terms:
```

Groups	Name	Std.Dev.
group	(Intercept)	0.2351

```
Num. levels: group 30
```

$$\rightarrow \hat{\mu}_a^{\text{mean}} = -0.812, \hat{\beta}^{\text{mean}} = 0.201, \hat{a}_1^{\text{mean}} = 0.002, \dots, \hat{a}_{30}^{\text{mean}} = 0.028, \hat{r}^{\text{mean}} = 1.927, \hat{\sigma}_a^{2,\text{mean}} = 0.055$$

$$\rightarrow \hat{\mu}_a^{\text{med}} = -0.809, \hat{\beta}^{\text{med}} = 0.202, \hat{r}^{\text{med}} = 1.736, \hat{\sigma}_a^{2,\text{med}} = 0.2351$$

- Varying intercept, with group-level predictors

$$\begin{cases} y_i^{\text{count}} \sim \text{NB}(\exp(a_{j[i]} + \beta x_i), r) \\ a_j \sim \mathcal{N}(\alpha_0 + \alpha_1 w_j, \sigma_a^2) \end{cases}$$

```
> m2 = rstanarm::stan_glmer.nb(y_count ~ 1 + x + w + (1 | group), mydata)
> rstanarm::summary(m2)
```

	mean	mcse	sd
(Intercept)	-0.834	0.005	0.108
x	0.170	0.003	0.094
w	0.171	0.003	0.075
b[(Intercept) group:1]	0.016	0.007	0.184
...			
b[(Intercept) group:30]	0.051	0.008	0.196
reciprocal_dispersion	2.058	0.034	0.866
Sigma[group:(Intercept),(Intercept)]	0.046	0.003	0.057
mean_PPD	0.473	0.002	0.063
log-posterior	-320.796	0.306	4.881

```
> rstanarm::print(m2)
```

	Median	MAD_SD
(Intercept)	-0.831	0.106

⁷⁷ `Δ` `rstanarm::summary()` shows posterior means and SDs, whereas `rstanarm::print()` and `broom::tidy()` show more robust estimators of the posterior distributions: posterior medians and MAD-SDs (estimates of the standard deviation of the marginal posterior distributions, based on a scaling of the Median Absolute Deviation—MAD—from the posterior medians).

```
x          0.173  0.095
w          0.172  0.074
```

```
Auxiliary parameter(s):
                        Median MAD_SD
reciprocal_dispersion 1.885  0.697
```

```
Error terms:
Groups Name      Std.Dev.
group (Intercept) 0.2155
Num. levels: group 30
```

$$\rightarrow \hat{\alpha}_0^{\text{mean}} = -0.834, \hat{\beta}^{\text{mean}} = 0.170, \hat{\alpha}_1^{\text{mean}} = 0.171, \hat{a}_1^{\text{mean}} = 0.016, \dots, \hat{a}_{30}^{\text{mean}} = 0.051, \hat{r}^{\text{mean}} = 2.058, \hat{\sigma}_a^{2,\text{mean}} = 0.046$$

$$\rightarrow \hat{\alpha}_0^{\text{med}} = -0.831, \hat{\beta}^{\text{med}} = 0.173, \hat{\alpha}_1^{\text{med}} = 0.172, \hat{r}^{\text{med}} = 1.885, \hat{\sigma}_a^{\text{med}} = 0.2155$$

- Co-varying intercept and slope, no group-level predictors

$$\begin{cases} y_i^{\text{count}} \sim \text{NB}(\exp(a_j[i] + b_j[i] x_i), r) \\ \begin{bmatrix} a_j \\ b_j \end{bmatrix} \sim \text{MVN}\left(\begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{bmatrix}\right) \end{cases}$$

```
> m3 = rstanarm::stan_glmer.nb(y_count ~ 1 + x + (1 + x | group), mydata)
> rstanarm::summary(m3)
```

```
              mean  mcse    sd
(Intercept) -0.802  0.004 0.112
x            0.195  0.003 0.102
b[(Intercept) group:1] -0.015 0.005 0.159
b[x group:1] -0.006 0.005 0.130
...
b[(Intercept) group:30] 0.016 0.005 0.154
b[x group:30] -0.017 0.004 0.114
reciprocal_dispersion 1.921 0.028 0.780
Sigma[group:(Intercept),(Intercept)] 0.032 0.003 0.048
Sigma[group:x,(Intercept)] -0.004 0.001 0.019
Sigma[group:x,x] 0.017 0.001 0.029
mean_PPD 0.472 0.002 0.068
log-posterior -371.125 0.435 6.543
```

```
> rstanarm::print(m3)
      Median MAD_SD
(Intercept) -0.797  0.110
x            0.193  0.099
```

```
Auxiliary parameter(s):
                        Median MAD_SD
reciprocal_dispersion 1.778  0.659
```

```
Error terms:
Groups Name      Std.Dev. Corr
group (Intercept) 0.1791
      x          0.1322  -0.16
Num. levels: group 30
```

$$\rightarrow \hat{\mu}_a^{\text{mean}} = -0.802, \hat{\mu}_b^{\text{mean}} = 0.195, \hat{a}_1^{\text{mean}} = -0.015, \hat{b}_1^{\text{mean}} = -0.006, \dots, \hat{a}_{30}^{\text{mean}} = 0.016, \hat{b}_{30}^{\text{mean}} = -0.017, \hat{r}^{\text{mean}} = 1.921, \hat{\sigma}_a^{2,\text{mean}} = 0.032, \hat{\sigma}_b^{2,\text{mean}} = 0.017, \hat{\sigma}_{ab}^{\text{mean}} = -0.004$$

$$\rightarrow \hat{\mu}_a^{\text{med}} = -0.797, \hat{\mu}_b^{\text{med}} = 0.193, \hat{r}^{\text{med}} = 1.778, \hat{\sigma}_a^{\text{med}} = 0.1791, \hat{\sigma}_b^{\text{med}} = 0.1322, \hat{\rho}_{ab}^{\text{med}} := \frac{\hat{\sigma}_{ab}}{\hat{\sigma}_a \hat{\sigma}_b} = -0.16$$

- Co-varying intercept and slope, with group-level predictors

$$\begin{cases} y_i^{\text{count}} \sim \text{NB}(\exp(a_j[i] + b_j[i] x_i), r) \\ \begin{bmatrix} a_j \\ b_j \end{bmatrix} \sim \text{MVN}\left(\begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix} + \begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix} w_j, \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{bmatrix}\right) \end{cases}$$

```

> m4 = rstanarm::stan_glmer.nb(y_count ~ 1 + x + w + x:w + (1 + x | group), mydata)
> rstanarm::summary(m4)

```

	mean	mcse	sd
(Intercept)	-1.127	0.011	0.225
x	0.315	0.012	0.254
w	0.163	0.009	0.200
x:w	-0.041	0.012	0.244
b[(Intercept) group:1]	-0.150	0.023	0.607
b[x group:1]	-0.733	0.025	0.651
...			
b[(Intercept) group:30]	-0.478	0.018	0.537
b[x group:30]	-0.754	0.021	0.624
reciprocal_dispersion	2.066	0.026	0.650
Sigma[group:(Intercept),(Intercept)]	0.789	0.022	0.411
Sigma[group:x,(Intercept)]	0.833	0.021	0.371
Sigma[group:x,x]	1.487	0.026	0.531
mean_PPD	1.157	0.009	0.252
log-posterior	-381.226	0.588	7.884

```

> rstanarm::print(m4)

```

	Median	MAD_SD
(Intercept)	-1.120	0.218
x	0.317	0.239
w	0.164	0.186
x:w	-0.042	0.234

Auxiliary parameter(s):

	Median	MAD_SD
reciprocal_dispersion	1.955	0.610

Error terms:

Groups	Name	Std.Dev.	Corr
group	(Intercept)	0.8884	
	x	1.2195	0.77

Num. levels: group 30

$$\begin{aligned}
& - \hat{\alpha}_0^{\text{mean}} = -1.127, \hat{\beta}_0^{\text{mean}} = 0.315, \hat{\alpha}_1^{\text{mean}} = 0.163, \hat{\beta}_1^{\text{mean}} = -0.041, \hat{a}_1^{\text{mean}} = -0.150, \hat{b}_1^{\text{mean}} = -0.733, \dots, \\
& \hat{a}_{30}^{\text{mean}} = -0.478, \hat{b}_{30}^{\text{mean}} = -0.754, \hat{r}^{\text{mean}} = 2.066, \hat{\sigma}_a^{2,\text{mean}} = 0.789, \hat{\sigma}_{ab}^{\text{mean}} = 0.833, \hat{\sigma}_b^{2,\text{mean}} = 1.487 \\
& - \hat{\alpha}_0^{\text{med}} = -1.120, \hat{\beta}_0^{\text{med}} = 0.317, \hat{\alpha}_1^{\text{med}} = 0.164, \hat{\beta}_1^{\text{med}} = -0.042, \hat{r}^{\text{med}} = 1.955, \hat{\sigma}_a^{\text{med}} = 0.8884, \hat{\sigma}_b^{\text{med}} = \\
& 1.2195, \hat{\rho}_{ab}^{\text{med}} := \frac{\hat{\sigma}_{ab}}{\hat{\sigma}_a \hat{\sigma}_b} = 0.77
\end{aligned}$$